

Oblate Spheroid Excitation Theory: A Unified, Lattice-Free Foundation for Plastic Deformation from Which Dislocations Emerge as Collective Excitations

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Abstract

Dislocation theory has underpinned crystal plasticity for a century, yet its lattice-dependent definition cannot describe plastic flow in grain boundaries, glasses, ceramics, or nanocrystals near the glass transition, where no periodic lattice exists. We propose the *Oblate Spheroid Excitation Theory* (OSET): the elementary carrier of plastic deformation, in any solid, is a shear-eigenstrained oblate spheroid, the oblate-spheroidal transformation zone (OSTZ), treated within Eshelby's inclusion theory. The OSTZ requires no lattice and has a finite, non-singular, intrinsically thermally activated energy and stress field. Three results are proved: a single OSTZ produces a non-singular elastic dipole, not a dislocation's singular field; a co-planar chain of N OSTZs is mathematically identical to a Peierls–Nabarro dislocation, core width and Burgers vector fixed by OSTZ geometry; and a genuine dislocation nucleates only once the chain reaches a host-lattice-set critical length. Dislocations emerge as a collective, large- N limit of OSET rather than an assumed entity, and the theoretical shear strength, Peierls stress, core energy, Frank–Read critical stress, and stacking-fault energy follow as derived, parameter-free quantities. OSET is validated against grain-boundary-sliding data, independent literature spanning metals, ceramics, and bulk metallic glasses, and a recent 41-system compilation, reproducing the fitted dilatational and shear eigenstrains to within 2% and 15%, respectively. Because classical dislocation theory emerges from OSET but OSET does not require dislocations, it provides, in our view, a more fundamental, broadly applicable foundation for plastic deformation across material classes.

Keywords: oblate spheroid excitation, Eshelby inclusion, dislocation emergence, grain-boundary sliding, Peierls–Nabarro model, superplasticity

1. Introduction

The dislocation is, by a wide margin, the most successful concept in the theory of slightly defective crystal plasticity. Since its independent introduction by Taylor, Orowan, and Polanyi in 1934 (Taylor, 1934; Orowan, 1934; Polanyi, 1934) to explain why real crystals yield at stresses orders of magnitude below the theoretical shear strength, the dislocation has provided the microscopic basis for strain hardening, the Hall–Petch average grain-size dependence of strength (Hall, 1951; Petch, 1953; Dunstan and Bushby, 2014; Zhang et al., 2020), creep, fatigue, and fracture in essentially every crystalline engineering alloy. Its mathematical foundation was laid by Volterra's elastic continuum construction of a line defect (Volterra, 1907) decades before its physical role was understood, and was subsequently refined by the Peierls–Nabarro model (Peierls, 1940; Nabarro, 1947), which regularised the otherwise singular core by spreading the displacement discontinuity over a finite width and remains an active area of core-structure modelling (Xu et al., 2020), and by Eshelby's general theory of elastic inclusions (Eshelby, 1957), which placed the energetics of lattice defects on a rigorous continuum-mechanical footing. Eight decades of refinement have made dislocation theory, enriched by

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kink-pair nucleation, climb, cross-slip, and forest-hardening models, the standard language in which crystal plasticity is presently taught, computed, and engineered (Hirth and Lothe, 1982; McDowell and Liu, 2025).

That success rests, however, on a definition that is fundamentally topological rather than physical. The dislocation is specified by the closure failure of a Burgers circuit traversed in a Bravais lattice,

$$\oint_C du_i = b_i, \quad b_i \in \Lambda_{\text{Bravais}}, \quad (1)$$

and this definition carries three restrictions that are unavoidable consequences of its topological character rather than incidental simplifications. **(R1)** It requires a crystal lattice: without a Bravais lattice there is no notion of a closed reference circuit, no quantised Burgers vector Λ_{Bravais} , and therefore no dislocation, so grain boundaries, glasses, and polymers are excluded from the description by definition rather than by approximation. **(R2)** Its core is singular: the Volterra stress field $\sigma \sim Gb/2\pi r$ diverges as $r \rightarrow 0$, forcing the introduction of an empirical cutoff radius $r_0 \sim b$ and a core energy that must be fitted to experiment or atomistic simulation rather than derived. **(R3)** It is athermal at the elementary level: the bare dislocation carries no intrinsic activation energy or entropy, so thermal activation has to be appended afterwards through kink-nucleation or Peierls–Nabarro overlays rather than emerging from the definition itself.

These three restrictions are not merely formal concerns; each one fails in a regime of substantial current interest. In grain-boundary sliding and superplastic deformation, the dislocation content of a boundary is conventionally described through DSC (displacement-shift-complete) lattice vectors, but for a general high-angle boundary the DSC lattice is commensurate with the two adjoining crystals only in a statistical sense, so the underlying reference lattice that R1 demands is simply not available; this is precisely the regime in which Padmanabhan and Schlipf (1996); Padmanabhan and Gleiter (2014); Padmanabhan and Divinski (2024) introduced a phenomenological rate equation, built around an empirical boundary-sliding eigenstrain, that successfully predicts superplastic flow without ever invoking a dislocation explicitly, a regime more recently also addressed by coupling grain rotation and dislocation plasticity directly to grain-boundary sliding (Borodin et al., 2020; Huang et al., 2022; Hua and Zhao, 2022; Huang et al., 2024; Du et al., 2025), and documented experimentally in nanocrystalline multiphase networks exhibiting anomalously low-temperature, high-strain-rate superplasticity (Wang et al., 2023; Zhang et al., 2023). In nanocrystalline metals with average grain size $d < 10b$, standard dislocation-based hardening theory predicts a continued increase in strength with decreasing grain size that is not observed experimentally; instead, below a critical size the flow stress falls with further grain refinement, the inverse Hall–Petch effect (Ovid’ko and Sheinerman, 2017; Li et al., 2024b; Han and Yi, 2022; Li et al., 2024a; Ma et al., 2024; Cappola et al., 2024). This softening was reported experimentally and given an early phenomenological explanation, in terms of a grain-boundary-sliding-controlled flow process, well before it was reproduced in atomistic simulation (Hahn and Padmanabhan, 1997; Schiøtz et al., 1998), because a grain too small to host an extended dislocation network cannot be described by a theory whose elementary unit is a dislocation line. In metallic glasses there is no lattice at all, and yet the material deforms plastically; the accepted microscopic picture is the shear-transformation zone (STZ) (Argon, 1979; Falk and Langer, 1998; Zhu et al., 2024), a local, thermally activated rearrangement of a cluster of atoms that bears no resemblance to a topological line defect. Ceramics and semicrystalline polymers present further intermediate cases: wherever dislocation glide is restricted to a small number of slip systems or absent altogether, the same absence of a usable reference lattice (R1) again forces boundary- and free-volume-mediated mechanisms to dominate the observed plasticity.

The existing remedies for these failures are mechanism-specific rather than unified. STZ theory (Argon, 1979; Falk and Langer, 1998) supplies a phenomenological activation volume and energy for amorphous flow but does not derive either from an explicit elastic inclusion, so it offers no route back to dislocation quantities such as the Peierls stress or stacking-fault energy. The grain-boundary sliding theory of Padmanabhan and Schlipf (1996) is, in the present context, the closest predecessor to this work: it identifies the elementary boundary-sliding event with an oblate-spheroidal region of grain boundary atomic ensembles and obtains a rate equation in excellent agreement with experiment, but its activation energy is expressed in terms of Eshelby constraint factors that were assumed rather than derived from the underlying I -integrals, and the theory was never connected back to the dislocation quantities of single-crystal plasticity. Composite models that partition a nanocrystal into grain-interior and grain-boundary-zone contributions (Barai and

Weng, 2009; Li et al., 2021) capture the same physical separation phenomenologically, again without a single elementary excitation common to both zones. The general idea that deformation in disordered or sub-dislocation-scale regions might require an elementary unit smaller than a lattice dislocation was raised by Padmanabhan and Gleiter (2012) in an earlier review and revisited in the context of grain/interphase boundary structure (Padmanabhan and Gleiter, 2014), but in neither case was the idea developed into a quantitative theory with derivable, parameter-free observables. What is missing, in short, is a single elementary excitation that (i) requires no crystal lattice for its definition, (ii) has a finite, non-singular core energy obtained from elasticity theory rather than fitted to data, (iii) is intrinsically thermally activated, and (iv) reduces to the conventional dislocation, with all of its derived properties, in the appropriate collective limit.

This paper proposes such an excitation: the *oblate spheroidal transformation zone* (OSTZ), and develops it into a complete framework that we call the *Oblate Spheroid Excitation Theory* (OSET). Physically, an OSTZ is the smallest unit of irreversible shear a solid can sustain without invoking a lattice: a disk-shaped region, of the same size as a free-volume cluster at a grain boundary or a shear-transformation zone in a glass, that flips into a sheared configuration when thermally or mechanically activated. Modelling this region as an Eshelby oblate spheroid, rather than as a sphere or a flat crack, is the key geometric choice of the theory: the shape is thin enough to localise shear near a glide plane, yet smooth enough that Eshelby’s inclusion theory (Eshelby, 1957; Mura, 1987) yields an exact, closed-form, and non-singular stress and energy. OSET requires exactly three material inputs, the OSTZ radius W , the shear eigenstrain γ_0 , and the shear modulus G of the host, and from these derives every other quantity used in this paper.

Three central results are established below. First, the far field of a single OSTZ is shown to be a stress *dipole* rather than a monopole (§3), decaying as r^{-3} and therefore incapable, on its own, of producing the long-range r^{-1} field of a dislocation. Second, a co-planar chain of N OSTZs is proved to be mathematically identical to a Peierls–Nabarro dislocation (§4), with core half-width $\zeta = W$ and Burgers vector $b = N\gamma_0 W$ following directly from the chain construction rather than being assumed. Third, a full lattice dislocation is shown to nucleate only once the chain reaches a critical length $N = N_c = b_{\text{latt}}/(\gamma_0 W)$, at which point the Burgers vector locks onto a Bravais lattice vector. Because the dislocation emerges from OSTZ mechanics, rather than the reverse, OSET further yields, as derived rather than fitted quantities, the theoretical shear strength, the Peierls stress, the dislocation core energy, the Frank–Read critical stress, and the stacking-fault energy (§5), together with a grain-boundary energy expression (§6) and a statistical-mechanical treatment of the OSTZ population that recovers the Padmanabhan et al. superplastic rate equation and a Taylor-hardening law (§7). Sections 9–11 extend the framework across material classes, state its novel testable predictions, and validate it quantitatively against the original 1996 Padmanabhan et al. data set, subsequent independent literature, and the 41-system compilation of Harisankar and Padmanabhan (2025).

2. The Fundamental Entity: the OSTZ

Physically, an OSTZ is the smallest unit of irreversible shear that a solid can sustain without invoking a crystal lattice: a roughly disk-shaped pocket of material, of the same size as a free-volume cluster at a grain boundary or a shear-transformation zone in a glass, that flips into a sheared configuration when thermally or mechanically activated. Modelling it as an *oblate spheroid* rather than a sphere or a flat crack is the key geometric choice of this paper: it is thin enough to represent a localized shear event confined near a boundary or free-volume site, yet smooth enough that Eshelby’s inclusion theory (Eshelby, 1957) gives an exact, closed-form, non-singular stress and energy. The next two subsections fix this geometry and then derive its energetics from first principles.

2.1. Geometric definition

Only two numbers are needed to fix the shape of the OSTZ: its in-plane radius W , set by the free-volume length scale of the host material, and its aspect ratio α , fixed once and for all by requiring the spheroid to be oblate enough to localise shear near a glide plane while remaining smooth enough for Eshelby’s exact

solution to apply. We adopt $\alpha = 1/2$ throughout: an OSTZ is an oblate spheroid that has undergone a shear eigenstrain, with semi-axes

$$\begin{aligned} a_1 = a_2 = W, \quad a_3 = W/2, \quad \alpha \equiv a_3/a_1 = \frac{1}{2}, \\ V_0 = \frac{4\pi}{3} a_1^2 a_3 = \frac{2\pi}{3} W^3. \end{aligned} \quad (2)$$

The flat face lies in the x_1 - x_2 glide plane, so that shear parallel to this plane corresponds to relative sliding of the two flat faces of the oblate spheroid past each other, the same kinematics as grain-boundary sliding or a shear-transformation event. The eigenstrain that produces this sliding, together with the accompanying volume change measured by ε_0 , is

$$\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2 \text{ (shear)}, \quad \varepsilon_{ii}^* = \varepsilon_0/3 \text{ (dilat.)} \quad (3)$$

Because the eigenstrain is a pure 13-shear rather than an in-plane (12) shear, this geometry makes S_{1313} , not S_{1212} , the Eshelby tensor component governing the stored elastic energy: the central quantity computed in the rest of this section.

2.2. Eshelby solution for the OSTZ

We now need the actual stress and strain that the eigenstrain of Eq. (3) produces inside the OSTZ, since this is what determines how much elastic energy is stored when the OSTZ activates (§2.4). For an arbitrary inclusion shape this would require solving an elasticity boundary-value problem numerically; the entire point of choosing an *ellipsoid* is that Eshelby's inclusion theorem (Eshelby, 1957) gives the answer in closed form. The theorem states that the strain and stress inside a homogeneous ellipsoidal inclusion are *uniform* (independent of position within the inclusion):

$$\varepsilon_{ij}^{\text{in}} = S_{ijkl} \varepsilon_{kl}^*, \quad (4)$$

$$\sigma_{ij}^{\text{in}} = C_{ijkl} (S_{klmn} - I_{klmn}) \varepsilon_{mn}^*. \quad (5)$$

All of the shape-dependence is packaged into one object, the Eshelby tensor S_{ijkl} ; uniformity of $\varepsilon_{ij}^{\text{in}}$ holds only for ellipsoids, because they are the only shapes for which S_{ijkl} does not vary from point to point inside the inclusion.

Since the OSTZ eigenstrain (Eq. (3)) is a pure 13-shear, only the single component S_{1313} is needed; for the general ellipsoid this component is built from two auxiliary shape integrals, I_α and $I_{\alpha\beta}$, that depend only on the semi-axes (Mura, 1987):

$$S_{1313} = \frac{(1 + \alpha^2)I_{13} + (1 - 2\nu)(I_1 + I_3)}{16\pi(1 - \nu)}, \quad (6)$$

where $\Delta(s) = \sqrt{(a_1^2 + s)(a_2^2 + s)(a_3^2 + s)}$,

$$\begin{aligned} I_\alpha &= 2\pi a_1 a_2 a_3 \int_0^\infty \frac{ds}{(a_\alpha^2 + s)\Delta(s)}, \\ I_{\alpha\beta} &= 2\pi a_1 a_2 a_3 \int_0^\infty \frac{ds}{(a_\alpha^2 + s)(a_\beta^2 + s)\Delta(s)}, \end{aligned} \quad (7)$$

with sum rule $I_1 + I_2 + I_3 = 4\pi$. These integrals are elementary but tedious; the next three subsections evaluate them explicitly for our oblate spheroid ($a_1 = a_2 = 1$, $a_3 = \alpha$ in reduced units), first deriving I_1 in closed form, then I_{13} , then substituting numbers for $\alpha = 1/2$ to obtain the constraint factor β_1 that ultimately enters the activation energy of Eq. (22).

2.2.1. Derivation of I_1

Starting point. Focus on $a_1 = a_2 = 1$, $a_3 = \alpha$. Then $\Delta(s) = (1 + s)\sqrt{\alpha^2 + s}$ and

$$I_1 = 2\pi\alpha \int_0^\infty \frac{ds}{(1 + s)^2 \sqrt{\alpha^2 + s}}. \quad (8)$$

Substitution. Let $t = \sqrt{\alpha^2 + s}$, so $s = t^2 - \alpha^2$, $ds = 2t dt$, $\sqrt{\alpha^2 + s} = t$, and $1 + s = t^2 + (1 - \alpha^2) \equiv t^2 + e^2$ (where $e^2 \equiv 1 - \alpha^2$). New limits: $t : \alpha \rightarrow \infty$. Substituting:

$$I_1 = 4\pi\alpha \int_{\alpha}^{\infty} \frac{dt}{(t^2 + e^2)^2}. \quad (9)$$

Antiderivative. The reduction formula (proved by differentiating the right side):

$$\int \frac{dt}{(t^2 + e^2)^2} = \frac{t}{2e^2(t^2 + e^2)} + \frac{\arctan(t/e)}{2e^3} + C. \quad (10)$$

Proof. Let $F = t/[2e^2(t^2 + e^2)]$, $G_f = \arctan(t/e)/(2e^3)$:

$$\begin{aligned} F' &= \frac{1}{2e^2} \cdot \frac{(t^2 + e^2) - t \cdot 2t}{(t^2 + e^2)^2} = \frac{e^2 - t^2}{2e^2(t^2 + e^2)^2}, \\ G'_f &= \frac{1}{2e^3} \cdot \frac{1/e}{1 + (t/e)^2} = \frac{1}{2e^2(t^2 + e^2)}. \end{aligned}$$

Sum: $F' + G'_f = [(e^2 - t^2) + (t^2 + e^2)]/[2e^2(t^2 + e^2)^2] = 1/(t^2 + e^2)^2$. $\square$

Evaluating at the bounds. Upper bound ($t \rightarrow \infty$): $F \rightarrow 0$, $\arctan \rightarrow \pi/2$, so $[F + G_f]_{t \rightarrow \infty} = \pi/(4e^3)$.

Lower bound ($t = \alpha$): note $\alpha^2 + e^2 = 1$, so $F|_{\alpha} = \alpha/(2e^2)$, $G_f|_{\alpha} = \arctan(\alpha/e)/(2e^3)$. Hence:

$$\begin{aligned} \int_{\alpha}^{\infty} \frac{dt}{(t^2 + e^2)^2} &= \frac{\pi}{4e^3} - \frac{\alpha}{2e^2} - \frac{\arctan(\alpha/e)}{2e^3} \\ &= \frac{1}{2e^3} \left[\frac{\pi}{2} - \arctan\left(\frac{\alpha}{e}\right) \right] - \frac{\alpha}{2e^2}. \end{aligned} \quad (11)$$

Identity $\pi/2 - \arctan(\alpha/e) = \arccos \alpha$. Set $\alpha = \cos \phi$ ($e = \sin \phi$). Then $\arctan(\cos \phi / \sin \phi) = \arctan(\cot \phi) = \pi/2 - \phi$, so $\pi/2 - \arctan(\alpha/e) = \phi = \arccos \alpha$. Substituting into Eq. (11) and multiplying by $4\pi\alpha$, and replacing $e^3 = (1 - \alpha^2)^{3/2}$:

$$\boxed{I_1 = \frac{2\pi\alpha}{(1 - \alpha^2)^{3/2}} \left[\arccos \alpha - \alpha\sqrt{1 - \alpha^2} \right], \quad I_3 = 4\pi - 2I_1.} \quad (12)$$

2.2.2. *Derivation of $I_{13} = (I_3 - I_1)/(1 - \alpha^2)$*

Equation (6) for S_{1313} still needs the cross-integral I_{13} , which superficially requires evaluating yet another integral of the same type as I_1 . In fact I_{13} can be obtained algebraically from I_1 and I_3 without any new integration, by exploiting the way the two denominators $(1 + s)$ and $(\alpha^2 + s)$ combine. Writing the integrals explicitly ($\Delta = (1 + s)\sqrt{\alpha^2 + s}$):

$$\begin{aligned} I_1 &= 2\pi\alpha \int_0^{\infty} \frac{ds}{(1 + s)^2 \sqrt{\alpha^2 + s}}, \\ I_3 &= 2\pi\alpha \int_0^{\infty} \frac{ds}{(1 + s)(\alpha^2 + s)^{3/2}}, \\ I_{13} &= 2\pi\alpha \int_0^{\infty} \frac{ds}{(1 + s)^2 (\alpha^2 + s)^{3/2}}. \end{aligned}$$

Forming $I_3 - I_1$ and factoring out $1/[(1 + s)\sqrt{\alpha^2 + s}]$:

$$I_3 - I_1 = 2\pi\alpha \int_0^{\infty} \frac{1}{(1 + s)\sqrt{\alpha^2 + s}} \left[\frac{1}{\alpha^2 + s} - \frac{1}{1 + s} \right] ds.$$

The bracket equals $(1 - \alpha^2)/[(\alpha^2 + s)(1 + s)]$:

$$\begin{aligned} I_3 - I_1 &= (1 - \alpha^2) \cdot 2\pi\alpha \int_0^\infty \frac{ds}{(1 + s)^2(\alpha^2 + s)^{3/2}} \\ &= (1 - \alpha^2)I_{13}. \end{aligned}$$

Therefore:

$$\boxed{I_{13} = \frac{I_3 - I_1}{1 - \alpha^2}} \quad \square \quad (13)$$

2.2.3. Numerical evaluation ($\alpha = 1/2$, $\nu = 1/3$)

With I_1 and I_{13} now available in closed form, we can finally put in the actual OSTZ aspect ratio $\alpha = 1/2$ and a representative Poisson ratio $\nu = 1/3$ to obtain the single number, β_1 , that controls the shear-energy cost of activating an OSTZ (Eq. (17) below).

Step 1: compute I_1 . $e = \sqrt{1 - 0.25} = 0.8660$. Bracket: $\arccos(0.5) - 0.5 \times 0.8660 = 1.0472 - 0.4330 = 0.6142$. $(0.75)^{3/2} = 0.6495$. $I_1 = \pi \times 0.6142/0.6495 = \mathbf{2.972}$.

Step 2: I_3 and I_{13} . $I_3 = 4\pi - 2 \times 2.972 = 12.566 - 5.944 = \mathbf{6.622}$. $I_{13} = (6.622 - 2.972)/0.75 = 3.650/0.75 = \mathbf{4.867}$.

Step 3: S_{1313} from Eq. (6). Numerator: $1.25 \times 4.867 + (1 - 2/3) \times 9.594 = 6.084 + 3.198 = 9.282$. Denominator: $16\pi \times (2/3) = 33.51$. $S_{1313} = 9.282/33.51 = \mathbf{0.2772}$.

Step 4: β_1 . $\beta_1 = 1 - 2 \times 0.2772 = \mathbf{0.4456} \approx 0.446$.

2.2.4. Limiting-case verification

Before trusting $\beta_1 = 0.446$ as a physically reasonable constraint factor, it is worth checking the formula against two shapes whose answer we already know by symmetry or by elementary physical reasoning: the sphere, where isotropy gives an independent check, and the vanishingly thin disk, which should behave like a stress-free crack.

Sphere ($\alpha \rightarrow 1$). For a sphere all semi-axes equal: $I_1 = I_2 = I_3 = 4\pi/3$. Direct integration: $I_{13}^{\text{sphere}} = 2\pi \int_0^\infty (1 + s)^{-7/2} ds = 2\pi[u^{-5/2}/(-5/2)]_1^\infty = 4\pi/5$. Substituting $(1 + \alpha^2 \rightarrow 2, I_1 + I_3 = 8\pi/3)$:

$$S_{1313}^{\text{sphere}} = \frac{4 - 5\nu}{15(1 - \nu)}. \quad (14)$$

For $\nu = 1/3$: $S_{1313}^{\text{sphere}} = 7/30 = 0.233$, $\beta_1^{\text{sphere}} = 8/15 \approx 0.533$.

Thin disk ($\alpha \rightarrow 0$). $I_3 \rightarrow 4\pi$, $I_{13} \rightarrow 4\pi$, $S_{1313}^{\text{disk}} = \frac{1}{2}$, $\beta_1 \rightarrow 0$: a crack-like disk stores no shear energy. The OSTZ aspect ratio $\alpha = 1/2$ gives the correct intermediate (Figure S1 in the Supporting Information plots $S_{1313}(\alpha)$ across this full range for several ν).

2.2.5. Dilatational constraint factor β_2

The shear eigenstrain is not the whole story: activating an OSTZ also requires a small volume change ε_0 (free-volume creation), and this dilatation costs its own elastic energy, governed by an analogous constraint factor β_2 . Because a volume change has no preferred direction, it is consistent to model this part of the problem with the simpler, fully isotropic spherical inclusion rather than repeating the oblate-spheroid calculation: for a spherical inclusion with purely dilatational eigenstrain $\varepsilon_{ij}^* = (\varepsilon_0/3)\delta_{ij}$, the sphere Eshelby tensor satisfies:

$$S_{ijkk}^{\text{sphere}} = \frac{1 + \nu}{3(1 - \nu)}\delta_{ij}. \quad (15)$$

This follows from the known sphere values (Mura, 1987) $S_{1111} = (7-5\nu)/[15(1-\nu)]$, $S_{1122} = (5\nu-1)/[15(1-\nu)]$, summing: $S_{1111} + S_{1122} + S_{1133} = (1+\nu)/[3(1-\nu)]$. Full derivation of β_2 follows in §2.4.2; the result:

$$\boxed{\beta_2 = \frac{4(1+\nu)}{9(1-\nu)}}. \quad (16)$$

For $\nu = 1/3$: $\beta_2 = 4(4/3)/[9(2/3)] = 0.889$.

Two formulas for β_2 exist: (a) exact oblate formula $\beta_2^{\text{oblate}} = 1 - S_{3333}(1-2\nu)/[2(1-\nu)] = 0.816$; (b) spherical dilatation approximation $\beta_2^{\text{sphere}} = 0.889$ (Padmanabhan and Schlipf, 1996). They differ by $\sim 9\%$ at $\nu = 1/3$; since $\beta_2 \varepsilon_0^2 \ll \beta_1 \gamma_0^2$, the difference is $< 2\%$ of ΔF_0 and inconsequential.

Table 1: Eshelby components for $\alpha = 1/2$.

ν	S_{1313}	S_{1212}	S_{3333}	β_1
0.30	0.2822	0.1769	0.7264	0.4356
1/3	0.2772	0.1739	0.7364	0.4456
0.35	0.2745	0.1723	0.7418	0.4510
0.40	0.2656	0.1670	0.7596	0.4688

2.3. Note on β_1 : two physical regimes

The value $\beta_1 \approx 0.45$ just derived is the purely elastic constraint factor for an OSTZ embedded in a matrix of its own shear modulus G . The original Padmanabhan et al. treatment of grain-boundary sliding, however, used an effective value $\beta_1^{\text{eff}} \approx 1$, more than double the elastic result. The two are not in conflict once the different elastic environment of a grain-boundary OSTZ is accounted for, as shown below.

(a) Eshelby (elastic) value: $\beta_1^{\text{Esh}} = 1 - 2S_{1313} \approx 0.44\text{--}0.49$ for $\alpha = 0.5$, $\nu = 0.28\text{--}0.44$.

(b) Padmanabhan et al. effective value: $\beta_1^{\text{eff}} \approx 1$.

Physical reconciliation. OSTZs at grain boundaries are embedded in GB material with $G_{\text{GB}} \approx G/2$ (GB softening (Wolf, 1990)): $\beta_1^{\text{eff}} = \beta_1^{\text{Esh}} \times G/G_{\text{GB}} \approx 0.45 \times 2 = 0.90 \approx 1$.

2.4. OSTZ elastic strain energy

We can now assemble the central result of this section: the activation energy ΔF_0 that a thermal fluctuation or applied stress must supply to flip a single OSTZ into its sheared configuration. Eshelby's theorem also gives a general formula for the elastic energy stored by an eigenstrained inclusion, as the work done by the (now known) internal stress against the eigenstrain: $\Delta F_0 = -\frac{1}{2} \sigma_{ij}^{\text{in}} \varepsilon_{ij}^* V_0$. Since the OSTZ eigenstrain has two physically distinct parts (the 13-shear of Eq. (3) and the dilatation ε_0), we evaluate their energy contributions separately below and add them at the end, justified by the fact that shear and dilatation are orthogonal modes of the isotropic stiffness tensor (no cross term). The dilatational contribution is the smaller of the two: for the canonical values $\gamma_0 = 0.1$, $\varepsilon_0 = 0.05$ it accounts for roughly 25% of the total ΔF_0 , but it is not negligible and is retained throughout.

2.4.1. Part 1: Shear contribution

Eigenstrain $\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2$, all other components are zero.

Step 1 (constrained strain). Eq. (4) gives $\varepsilon_{13}^{\text{in}}$ as a contraction of the Eshelby tensor with the *full* eigenstrain tensor, so both nonzero components ε_{13}^* and ε_{31}^* contribute: $\varepsilon_{13}^{\text{in}} = S_{1313}\varepsilon_{13}^* + S_{1331}\varepsilon_{31}^*$. By the minor symmetry of the Eshelby tensor, $S_{1313} = S_{1331}$ (a direct consequence of the symmetry $\varepsilon_{kl}^* = \varepsilon_{lk}^*$ of any physical strain tensor), so the two terms combine: $\varepsilon_{13}^{\text{in}} = S_{1313}(\gamma_0/2 + \gamma_0/2) = S_{1313}\gamma_0$.

Step 2 (elastic strain). The eigenstrain ε_{13}^* is, by definition, a stress-free transformation strain; it is the additional strain $e_{13}^{\text{el}} = \varepsilon_{13}^{\text{in}} - \varepsilon_{13}^*$, the difference between the actual (constrained) strain and the eigenstrain, that the surrounding matrix elastically resists and that therefore generates stress via Hooke's law: $e_{13}^{\text{el}} = \varepsilon_{13}^{\text{in}} - \varepsilon_{13}^* = (S_{1313} - \frac{1}{2})\gamma_0$.

Step 3 (interior stress). For an isotropic matrix the relevant stiffness component is simply twice the shear modulus, $C_{1313} = C_{1331} = G$ (the standard isotropic-elasticity relation between shear stress and shear strain), so $\sigma_{13}^{\text{in}} = 2Ge_{13}^{\text{el}} = 2G(S_{1313} - \frac{1}{2})\gamma_0$.

Step 4 (energy contraction). Both (1, 3) and (3, 1) contribute equally:

$$\Delta F_0^{\text{shear}} = -\frac{1}{2}(\sigma_{13}^{\text{in}}\varepsilon_{13}^* + \sigma_{31}^{\text{in}}\varepsilon_{31}^*)V_0 = -\frac{1}{2}\sigma_{13}^{\text{in}}\gamma_0 V_0.$$

Substituting: $\Delta F_0^{\text{shear}} = -\frac{1}{2} \cdot 2G(S_{1313} - \frac{1}{2})\gamma_0 \cdot \gamma_0 V_0 = G(\frac{1}{2} - S_{1313})\gamma_0^2 V_0$.

$$\boxed{\Delta F_0^{\text{shear}} = \frac{1}{2}\beta_1 G\gamma_0^2 V_0, \quad \beta_1 \equiv 1 - 2S_{1313}. \quad (17)}$$

Since $S_{1313} < \frac{1}{2}$ for any physical oblate spheroid, $\beta_1 > 0$ and the shear energy is positive definite.

2.4.2. Part 2: Dilatational contribution

The same energy bookkeeping is now repeated for the volumetric part of the eigenstrain, using the spherical-inclusion Eshelby result S_{ijkk}^{sphere} already obtained in §2.2.5 (Eq. (15)). Eigenstrain $\varepsilon_{ij}^* = (\varepsilon_0/3)\delta_{ij}$.

Step 1 (constrained strain inside a sphere). From Eqs. (4) and (15):

$$\varepsilon_{ij}^{\text{in}} = S_{ijkl}\varepsilon_{kl}^* = \frac{\varepsilon_0}{3} \cdot \frac{1+\nu}{3(1-\nu)}\delta_{ij} = \frac{(1+\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}. \quad (18)$$

Step 2 (elastic strain).

$$\begin{aligned} e_{ij}^{\text{el}} &= \varepsilon_{ij}^{\text{in}} - \varepsilon_{ij}^* = \varepsilon_0 \left[\frac{1+\nu}{9(1-\nu)} - \frac{1}{3} \right] \delta_{ij} \\ &= \varepsilon_0 \cdot \frac{(1+\nu) - 3(1-\nu)}{9(1-\nu)}\delta_{ij} = -\frac{2(1-2\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}. \end{aligned} \quad (19)$$

(Numerator: $1+\nu-3+3\nu=4\nu-2=-2(1-2\nu)$.) Volumetric trace: $e_{kk}^{\text{el}} = -6(1-2\nu)\varepsilon_0/[9(1-\nu)] = -2(1-2\nu)\varepsilon_0/[3(1-\nu)]$.

Step 3 (interior stress). Hooke's law for an isotropic solid splits the stress into a hydrostatic part proportional to the volumetric strain e_{kk}^{el} (governed by the Lamé constant λ) and a deviatoric part proportional to the local strain component itself (governed by $2G$); both parts must be added to get the total interior stress. Using $\lambda = 2G\nu/(1-2\nu)$:

$$\begin{aligned} \lambda e_{kk}^{\text{el}} &= \frac{2G\nu}{1-2\nu} \cdot \frac{-2(1-2\nu)\varepsilon_0}{3(1-\nu)} = -\frac{4G\nu\varepsilon_0}{3(1-\nu)}, \\ 2Ge_{11}^{\text{el}} &= -\frac{4G(1-2\nu)\varepsilon_0}{9(1-\nu)}. \end{aligned}$$

Adding the hydrostatic and deviatoric parts, with common denominator $9(1-\nu)$:

$$\begin{aligned} \sigma_{11}^{\text{in}} &= -\frac{12G\nu\varepsilon_0}{9(1-\nu)} - \frac{4G(1-2\nu)\varepsilon_0}{9(1-\nu)} \\ &= -\frac{G\varepsilon_0}{9(1-\nu)}[12\nu+4-8\nu] = -\frac{4G(1+\nu)\varepsilon_0}{9(1-\nu)}. \end{aligned} \quad (20)$$

Step 4 (energy contraction). By isotropy all three diagonal stress components are equal, $\sigma_{11}^{\text{in}} = \sigma_{22}^{\text{in}} = \sigma_{33}^{\text{in}}$, so the double contraction with the diagonal eigenstrain $\varepsilon_{ii}^* = \varepsilon_0/3$ collapses to three identical terms: $\sigma_{ij}^{\text{in}} \varepsilon_{ij}^* = 3\sigma_{11}^{\text{in}}(\varepsilon_0/3) = \sigma_{11}^{\text{in}}\varepsilon_0$. Substituting into the energy theorem,

$$\Delta F_0^{\text{dilat}} = \frac{1}{2}\beta_2 G \varepsilon_0^2 V_0, \quad \beta_2 = \frac{4(1+\nu)}{9(1-\nu)}. \quad (21)$$

Like β_1 , the dilatational constraint factor β_2 is strictly between 0 and 1, reflecting that the surrounding matrix relaxes part of the volumetric eigenstrain rather than resisting it completely; $\beta_2 = 0.889$ at $\nu = 1/3$ is closer to unity than $\beta_1 = 0.446$ because a spherical (volumetric) misfit is accommodated less efficiently by the elastic matrix than a shear misfit is.

2.4.3. Total activation energy

Shear and dilatational eigenstrains involve orthogonal tensor components (ε_{13}^* vs. ε_{ii}^*); their cross-terms vanish upon contraction with the isotropic stiffness tensor, so the two energies computed separately above can simply be added:

$$\Delta F_0 = \frac{1}{2}(\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2) G V_0, \quad V_0 = \frac{2\pi}{3} W^3. \quad (22)$$

The two constraint factors β_1 and β_2 together encode the resistance of the surrounding elastic matrix to the OSTZ shear and volume change respectively; both are less than 1 because the matrix partially accommodates each eigenstrain rather than resisting it as a rigid, undeformable cage, which is why the stored energy is always less than that of a hypothetical free (unconstrained) transformation.

Numerical check. This first-principles elastic result can be checked directly against the empirically calibrated activation energy used in the original Padmanabhan et al. grain-boundary sliding theory. Using the canonical grain-boundary values $\gamma_0 = 0.10$, $\varepsilon_0 = 0.05$, $\beta_1 = 0.446$, $\beta_2 = 0.889$, $W = 2.5b = 0.75$ nm (so $V_0 = 0.884$ nm³), and $G = 2.2 \times 10^{10}$ Pa for Al: $\Delta F_0 = \frac{1}{2}(0.00446 + 0.00222) \times 2.2 \times 10^{10} \times 0.884 \times 10^{-27} = 6.5 \times 10^{-20}$ J = 0.406 eV, within 6% of the calibrated value ≈ 0.38 eV, confirming that the Eshelby-derived constraint factors reproduce the empirically known activation energy without any further adjustment.

Table 2: Canonical OSTZ parameters.

Parameter	Symbol	GB/superpl.	Crystal
OSTZ radius	W	$2.5b$	$\approx b$
Shear eigenstrain	γ_0	0.10	s_m (0.10–0.20)
Dilat. eigenstrain	ε_0	0.05	—
Shear constraint	β_1	0.446	≈ 1
Dilat. constraint	β_2	0.889	—
Activation energy	ΔF_0	0.38 eV	—
Critical number	N_c	4	≈ 5 –10

3. Far-Field Stress: The Dipole Structure

Having fixed the energetics of a single OSTZ, we now ask what it looks like from far away, i.e. how it perturbs the stress in the surrounding grain. This is the question a materials scientist asks when picturing how one local shear event “talks to” its neighbours: does it act like a point source of stress (as a dislocation does), or is its influence short-ranged? The answer, derived below, is that a single OSTZ is a stress *dipole*, not a monopole: its field decays two powers of r faster than a dislocation’s. This is the microscopic reason a lone shear-transformation or grain-boundary sliding event cannot by itself propagate slip across a grain, and it sets up the chain construction of §4 in which many such dipoles must cooperate to reproduce a dislocation.

3.1. Why a dipole, not a monopole

The force monopole vanishes:

$$F_k = \oint_{\partial V_0} \sigma_{ij}^* n_j dS = C_{ijkl} \varepsilon_{kl}^* \underbrace{\oint_{\partial V_0} n_j dS}_{=0} = 0. \quad (23)$$

(The surface integral of the outward normal over any closed surface vanishes by the divergence theorem.) The leading far-field term is therefore the *force-dipole* at order r^{-3} .

The full multipole expansion (Mura, 1987) is

$$\sigma_{ij}(\mathbf{r}) = \underbrace{F_k \mathcal{G}_{ijk}}_{\text{monopole } r^{-2}} + \underbrace{M_{kl} \partial_l \mathcal{G}_{ijk}}_{\text{dipole } r^{-3}} + \underbrace{Q_{klm} \partial_l \partial_m \mathcal{G}_{ijk}}_{r^{-4}} + \dots \quad (24)$$

Since $F_k = 0$, the leading nonzero term is the dipole. The moment tensor (Mura, 1987) is

$$M_{kl} = C_{ijkl} \varepsilon_{ij}^* V_0 \Rightarrow M_{13} = M_{31} = G\gamma_0 V_0, \quad (25)$$

describing a double couple: two force pairs of equal and opposite sign, separated by the OSTZ diameter $2W$.

3.2. Step A: Kelvin Green's function and its derivative

The elastic displacement due to a unit point force at the origin in an infinite isotropic medium (Kelvin–Somigliana solution (Mura, 1987)):

$$\mathcal{G}_{ij}(\mathbf{r}) = A \left[\frac{(3-4\nu)\delta_{ij}}{r} + \frac{x_i x_j}{r^3} \right], \quad A = \frac{1}{16\pi G(1-\nu)}. \quad (26)$$

Differentiating the first term. Since $\partial_l(1/r) = -x_l/r^3$:

$$\partial_l \left[\frac{(3-4\nu)\delta_{ij}}{r} \right] = -\frac{(3-4\nu)\delta_{ij} x_l}{r^3}.$$

Differentiating the second term (product rule). $\partial x_i / \partial x_l = \delta_{il}$, $\partial(r^{-3}) / \partial x_l = -3x_l/r^5$:

$$\partial_l \left(\frac{x_i x_j}{r^3} \right) = \frac{\delta_{il} x_j + \delta_{jl} x_i}{r^3} - \frac{3x_l x_i x_j}{r^5}.$$

Combined result.

$$\frac{\partial \mathcal{G}_{ij}}{\partial x_l} = A \left[-\frac{(3-4\nu)\delta_{ij} x_l}{r^3} + \frac{\delta_{il} x_j + \delta_{jl} x_i}{r^3} - \frac{3x_l x_i x_j}{r^5} \right]. \quad (27)$$

3.3. Step B: Displacement from the moment tensor

For an Eshelby eigenstrain inclusion, the far-field exterior displacement is (Eshelby, 1957; Mura, 1987) $u_i = -M_{kl} \partial_l \mathcal{G}_{ik}$. With $M_{13} = M_{31} = G\gamma_0 V_0$ (all other components zero), the only nonzero contributions are $(k, l) = (1, 3)$ and $(3, 1)$.

Substituting Eq. (27) (with $\delta_{13} = 0$):

$$\begin{aligned} \frac{\partial \mathcal{G}_{i1}}{\partial x_3} &= A \left[-\frac{(3-4\nu)\delta_{i1} x_3}{r^3} + \frac{\delta_{i3} x_1}{r^3} - \frac{3x_3 x_i x_1}{r^5} \right], \\ \frac{\partial \mathcal{G}_{i3}}{\partial x_1} &= A \left[-\frac{(3-4\nu)\delta_{i3} x_1}{r^3} + \frac{\delta_{i1} x_3}{r^3} - \frac{3x_1 x_i x_3}{r^5} \right]. \end{aligned}$$

Adding term by term, the $\delta_{i1}x_3$ coefficient is $1 - (3 - 4\nu) = 4\nu - 2 = 2(2\nu - 1)$ (same for $\delta_{i3}x_1$), and the r^{-5} terms add to give $-6Ax_1x_3x_i/r^5$:

$$u_i = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(\delta_{i1}x_3 + \delta_{i3}x_1)}{r^3} - \frac{6x_1x_3x_i}{r^5} \right]. \quad (28)$$

Components relevant to σ_{13} :

$$u_1 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)x_3}{r^3} - \frac{6x_1^2x_3}{r^5} \right], \quad (29)$$

$$u_3 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)x_1}{r^3} - \frac{6x_1x_3^2}{r^5} \right]. \quad (30)$$

3.4. Step C: Stress from displacement gradients

Step B gave the displacement field $u_i(\mathbf{r})$ produced by the OSTZ dipole; what we actually need for the far-field stress (Eq. (35) below) is the shear stress σ_{13} , which by Hooke's law is built from the symmetric displacement gradient $\partial_3u_1 + \partial_1u_3$. Because the shear component has $\delta_{13} = 0$, the λ -term (the dilatational part of Hooke's law) drops out automatically, leaving a pure shear-modulus relation: $\sigma_{13} = G(\partial_3u_1 + \partial_1u_3)$. The remainder of this subsection differentiates the two terms of u_1 and u_3 found in Step B, term by term, and adds the results.

Auxiliary results. $\partial_3(x_3/r^3) = (r^2 - 3x_3^2)/r^5$; $\partial_3(x_3/r^5) = (r^2 - 5x_3^2)/r^7$; likewise for x_1 .

Computing ∂_3u_1 . Differentiating Eq. (29) with respect to x_3 :

$$\partial_3u_1 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(r^2 - 3x_3^2)}{r^5} - \frac{6x_1^2(r^2 - 5x_3^2)}{r^7} \right]. \quad (31)$$

Computing ∂_1u_3 . By the same procedure (swap $x_1 \leftrightarrow x_3$):

$$\partial_1u_3 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(r^2 - 3x_1^2)}{r^5} - \frac{6x_3^2(r^2 - 5x_1^2)}{r^7} \right]. \quad (32)$$

Summing and converting to unit vectors $\hat{r}_i = x_i/r$. $1/r^5$ group: $2(2\nu - 1)[2 - 3(\hat{r}_1^2 + \hat{r}_3^2)]/r^3$. $1/r^7$ group, expanding $(x_1^2 + x_3^2)r^2 - 10x_1^2x_3^2$: $[-6(\hat{r}_1^2 + \hat{r}_3^2) + 60\hat{r}_1^2\hat{r}_3^2]/r^3$. Collecting the $(\hat{r}_1^2 + \hat{r}_3^2)$ terms: $-6(2\nu - 1) - 6 = -12\nu$ (verify: $12\nu - 6 + 6 = 12\nu$). Hence:

$$\partial_3u_1 + \partial_1u_3 = -\frac{G\gamma_0 V_0 A}{r^3} [4(2\nu - 1) - 12\nu(\hat{r}_1^2 + \hat{r}_3^2) + 60\hat{r}_1^2\hat{r}_3^2]. \quad (33)$$

3.5. Step D: Angular tensor and far-field formula

Step C produced $\partial_3u_1 + \partial_1u_3$ as a function of position; multiplying by G converts it into the physical shear stress σ_{13} , and substituting the explicit prefactor $A = 1/[16\pi G(1 - \nu)]$ reduces the result to a single radial factor $1/r^3$ times a dimensionless function of direction only. That dimensionless function, denoted $\mathcal{T}_{13}(\hat{\mathbf{r}})$ below, is the *angular tensor*: it encodes how strongly the OSTZ promotes shear stress as a function of the direction $\hat{\mathbf{r}}$ from the source, independent of distance, and its sign and zeros (examined in Step E) determine where the OSTZ assists or opposes further slip. Multiplying Eq. (33) by G and inserting $A = 1/[16\pi G(1 - \nu)]$:

$$\sigma_{13} = \frac{G\gamma_0 V_0}{16\pi(1 - \nu)r^3} [-4(2\nu - 1) + 12\nu(\hat{r}_1^2 + \hat{r}_3^2) - 60\hat{r}_1^2\hat{r}_3^2].$$

Writing this as $G\gamma_0 V_0 \mathcal{T}_{13}/[2\pi(1 - \nu)r^3]$ requires dividing the bracket by 8 (ratio of prefactors $2\pi/16\pi = 1/8$):

$$\boxed{\mathcal{T}_{13}(\hat{\mathbf{r}}) = \frac{1}{2} - \nu + \frac{3\nu}{2}(\hat{r}_1^2 + \hat{r}_3^2) - \frac{15}{2}\hat{r}_1^2\hat{r}_3^2.} \quad (34)$$

$$\sigma_{13}(\mathbf{r}) = \frac{G\gamma_0 V_0}{2\pi(1-\nu)} \cdot \frac{\mathcal{T}_{13}(\hat{\mathbf{r}})}{r^3}. \quad (35)$$

The r^{-3} decay follows from differentiating the Kelvin r^{-1} Green's function twice: two orders faster than the Volterra r^{-1} field.

3.6. Step E: Verification in special directions

Equation (35) was derived through a long chain of differentiations, so before using it we check it in three special directions where the answer can be anticipated on physical grounds: the glide plane itself (where the formula must reduce to the expression used later in §3.7 and §4.1 to build up dislocation chains), the boundary-normal axis (where axial symmetry of the oblate spheroid demands a specific relation between directions), and the question of whether $\mathcal{T}_{13}$ ever changes sign (since a sign change would mean the OSTZ opposes slip in some directions, undermining its role as a cooperative nucleus).

Glide plane ($x_3 = 0, \hat{r}_3 = 0$). $\mathcal{T}_{13}|_{\hat{r}_3=0} = \frac{1}{2} - \nu + (3\nu/2)\hat{r}_1^2$. Expanding $r^2 = x_1^2 + x_2^2$ and collecting:

$$\sigma_{13}(x_1, x_2, 0) = \frac{G\gamma_0 V_0}{4\pi(1-\nu)r^5} [(1+\nu)x_1^2 + (1-2\nu)x_2^2]. \quad (36)$$

Positive definite for $0 < \nu < \frac{1}{2}$.

On-axis ($\hat{r}_1 = 0, \hat{r}_3 = 1$). $\mathcal{T}_{13} = \frac{1}{2} - \nu + 3\nu/2 = (1+\nu)/2 > 0$. By oblate axial symmetry, the same result holds along $\hat{x}_1$.

Nodal surface. Setting $\mathcal{T}_{13} = 0$ yields no real solution for $0 < \nu < \frac{1}{2}$: there is no nodal surface in the half-space $x_3 \geq 0$. The OSTZ promotes shear in all glide-plane directions (cooperative nucleus).

3.7. Glide-plane stress: six-step derivation

Step E confirmed the general 3D formula in special directions, but the calculation that matters for everything that follows, in particular the chain construction of §4, is the stress restricted to the glide plane itself, since that is where neighbouring OSTZs actually sit. Rather than simply substituting $x_3 = 0$ into Eq. (35) and Eq. (34) (which is algebraically awkward because of the way $\hat{r}_1$ and $\hat{r}_3$ enter), it is more transparent to recompute $\sigma_{13}|_{z=0}$ directly from the Green's function components, working through the off-diagonal and diagonal contributions one at a time and only afterwards converting to the unit-vector form. The six steps below carry this out and then re-express the result in terms of the effective Burgers vector b_{eff} (§3.8) so that it can be used directly as the building block for an OSTZ chain. Working at $x_3 = 0$ ($r^2 = x^2 + y^2, x \equiv x_1, y \equiv x_2$).

Step 1 (off-diagonal $\mathcal{G}_{13}$). $\partial^2(\mathcal{G}_{13})/(\partial x_1 \partial x_3)$ at $z = 0$: gives $(y^2 - 2x^2)/r^5$ (using $\partial_3(x_1 x_3/r^3)|_{z=0} = x_1/r^3$, then ∂_1).

Step 2 (diagonal $\mathcal{G}_{33}$). At $z = 0$: $\mathcal{G}_{33}|_{z=0} = A(3 - 4\nu)/r$. $\partial_1^2(1/r) = (3x^2 - r^2)/r^5 = (2x^2 - y^2)/r^5$:

$$\partial_1^2 \mathcal{G}_{33}|_{z=0} = A(3 - 4\nu)(2x^2 - y^2)/r^5.$$

Step 3 (diagonal $\mathcal{G}_{11}$). At $z = 0$: $\mathcal{G}_{11}|_{z=0} = A[(3 - 4\nu)/r + x^2/r^3]$. $\partial_3^2(1/r)|_{z=0} = -1/r^3$; $\partial_3^2(1/r^3)|_{z=0} = -3/r^5$:

$$\partial_3^2 \mathcal{G}_{11}|_{z=0} = A[-(3 - 4\nu)/r^3 - 3x^2/r^5].$$

Expanding: $-(3 - 4\nu)r^2 - 3x^2 = -(6 - 4\nu)x^2 - (3 - 4\nu)y^2$.

Step 4 (collect numerator, units of A/r^5).

$$\begin{aligned} x^2 \text{ coeff.: } 2(y^2 - 2x^2) &\rightarrow -4 \\ &+ (6 - 8\nu) - (3 - 4\nu) - 3 = -4(1 + \nu), \\ y^2 \text{ coeff.: } 2 - (3 - 4\nu) - (3 - 4\nu) &= -4(1 - 2\nu). \end{aligned}$$

Numerator: $-4A[(1 + \nu)x^2 + (1 - 2\nu)y^2]/r^5$.

Step 5 (assemble).

$$\sigma_{13}(x, y, 0) = \frac{G\gamma_0 V_0 [(1 + \nu)x^2 + (1 - 2\nu)y^2]}{4\pi(1 - \nu)r^5}.$$

Step 6 (re-express with b_{eff} , restore finite W). Using $b_{\text{eff}} = \gamma_0 W$:

$$\begin{aligned} G\gamma_0 V_0 &= G\gamma_0 \frac{2\pi W^3}{3} = Gb_{\text{eff}} \frac{2\pi W^2}{3}, \\ \text{so } \frac{G\gamma_0 V_0}{4\pi(1 - \nu)} &= \frac{Gb_{\text{eff}} W^2}{6(1 - \nu)}. \end{aligned}$$

Regularising $r^2 \rightarrow r^2 + W^2$:

$$\boxed{\sigma_{13}(x, y, 0) = \frac{Gb_{\text{eff}} W^2 [(1 + \nu)x^2 + (1 - 2\nu)y^2]}{6(1 - \nu)(x^2 + y^2 + W^2)^{5/2}}.} \quad (37)$$

Positive definite; finite at the origin.

Physical interpretation. Along $\hat{x}$ ($y = 0$): promotes forward shear. Along $\hat{y}$ ($x = 0$): promotes lateral shear. No nodal planes in the glide plane: the OSTZ is a *cooperative nucleus*.

3.8. Effective Burgers vector

The glide-plane formula just derived, Eq. (37), already anticipated that the natural strength parameter for an OSTZ is not γ_0 alone but a length-scaled combination b_{eff} with the same units as a dislocation Burgers vector; this is the quantity that will later sum, N at a time, into the total Burgers vector of an OSTZ chain (§4.2). Three independent routes to b_{eff} are checked against each other below; they agree to within $O(1)$, and the discrepancy itself is informative about the approximations involved.

Method 1 (kinematic, exact). The eigenstrain $\varepsilon_{13}^* = \gamma_0/2$ is, by definition, half the engineering shear strain $\gamma_0 = \partial u_1 / \partial x_3$ relating the relative glide displacement u_1 to the through-thickness coordinate x_3 . The total relative displacement between the two flat faces of the spheroid, separated by the full thickness $2a_3 = W$, is obtained by integrating this constant strain across the thickness:

$$b_{\text{eff}} = \int_{-a_3}^{a_3} \gamma_0 dx_3 = \gamma_0 \times 2a_3 = \gamma_0 W.$$

This is an exact kinematic statement, independent of the elastic solution, and is adopted as the primary definition.

Method 2 (moment tensor). As an independent check, b_{eff} can instead be extracted from the seismic moment $M_{13} = G\gamma_0 V_0$ of Eq. (25) by treating the OSTZ as a thin disk source of area πW^2 carrying a uniform Burgers vector b_{eff} , so that $M_{13} = Gb_{\text{eff}} \pi W^2$. Equating the two expressions for M_{13} and substituting $V_0 = (2\pi/3)W^3$:

$$\begin{aligned} Gb_{\text{eff}} \pi W^2 &= G\gamma_0 V_0 = G\gamma_0 \cdot \frac{2\pi W^3}{3} \\ \Rightarrow b_{\text{eff}} &= \frac{\gamma_0 \cdot \frac{2\pi W^3}{3}}{\pi W^2} = \frac{2\gamma_0 W}{3}, \end{aligned}$$

which underestimates Method 1 by 33%.

Origin of the discrepancy. The seismic-moment formula equates M_{13} to a Burgers vector spread uniformly over the disk area πW^2 , using the *full* oblate-spheroid volume $V_0 = (2\pi/3)W^3$ rather than the actual projected slip area. For a thin oblate disk ($\alpha \ll 1$) Methods 1 and 2 converge; at $\alpha = 1/2$ the 33% gap is non-negligible and is carried forward as an $O(1)$ uncertainty rather than resolved.

Method 3 (Lorentzian normalisation). The third, weaker check comes from the Lorentzian dislocation density derived later in §4.1, $\rho_1(x) = (b_{\text{eff}}/\pi)W/(x^2 + W^2)$ (Eq. (43)), which integrates to exactly b_{eff} over all x . This, however, holds *by construction*: the Lorentzian’s prefactor is chosen precisely so that the normalisation comes out to b_{eff} , so Method 3 cannot serve as an independent confirmation of the value found in Methods 1 and 2; it merely shows that the later construction is self-consistent with whichever b_{eff} is fed into it.

Method 1 is adopted as the primary definition throughout the rest of the paper:

$$\boxed{b_{\text{eff}} = \gamma_0 W.} \quad (38)$$

Table 3: Single OSTZ vs. Volterra dislocation.

Property	Single OSTZ	Volterra
Stress decay	r^{-3} (dipole)	r^{-1} (monopole)
Topological charge	0 (neutral)	$\pm b$
Net b	0	b
Core stress	Finite ($\leq G\gamma_0$)	$\rightarrow \infty$
Influence	$\sim W$	∞
Elastic E	$\frac{1}{2}\beta_1\gamma_0^2GV_0$	$Gb^2L\ln(R/r_0)$
Thermal activation	Intrinsic	Appended

3.9. Why a single OSTZ cannot drive macroscopic slip

The dipole structure (r^{-3} decay, Eq. (35)) has a decisive consequence: a single OSTZ cannot drive long-range slip. Its stress influence radius is effectively $\sim W$; beyond $\sim 5W$ the stress has fallen below 1% of its peak value. Only when $N = N_c$ OSTZs cooperate does the collective Burgers circuit close on a quantised $b \in \Lambda_{\text{Bravais}}$ (§4.4), leading to the long-range r^{-1} field of a full dislocation (Eq. (49)). Table 3 summarises the contrast.

4. Emergence of Dislocations from OSTZ Chains

This section answers the question raised in §3.9: how does a row of short-ranged dipoles add up to the long-range, singular field of a dislocation? We proceed in the same way a materials scientist would build up a slipped region from individual atomic-scale events: line up N OSTZs along the glide direction, sum their fields, and ask what collective object emerges. The result is exact and structurally important: an aligned OSTZ chain is mathematically identical to a Peierls–Nabarro dislocation, with the OSTZ radius W playing the role of the phenomenological core width that is normally fitted to experiment.

4.1. Lorentzian dislocation density: four steps

4.1.1. Step A: 2D dislocation density (BCS compliance)

The Bilby–Cottrell–Swinden (BCS) compliance (Bilby et al., 1963) $\delta u_1 = 2(1 - \nu)\sigma_{13}/G$ relates local slip to traction.

Physical basis. A shear traction σ_{13} on the glide plane produces a local slip increment δu_1 smaller than the naive estimate σ_{13}/G by a factor $(1 - \nu)$, because the surrounding half-space elastically constrains the opening of the slip increment in Poisson-ratio fashion; the factor can be obtained from the J-integral analysis of a shear crack (Rice, 1968) or, equivalently, from the dislocation-density relation $\sigma = G\rho/[2(1 - \nu)]$ of the Peierls–Nabarro model (Hirth and Lothe, 1982).

Applicability to the OSTZ r^{-3} field. Three features justify using the planar BCS compliance here even though it was derived for smooth slip distributions: (i) the OSTZ field decays as r^{-3} , faster than the dislocation’s r^{-1} , so the compliance integral converges absolutely without a short-distance cutoff; (ii) for $N \gg 1$ OSTZs the collective field is smooth on scales $\gg W$ and approaches the Peierls–Nabarro continuum limit; (iii) for grain-boundary OSTZs ($W = 2.5b$) the continuum approximation is well justified, while for crystal-interior OSTZs ($W \approx b$) it is marginal and atomistic corrections of $O(b/W)$ may be non-negligible.

Why an isotropic approximation is needed. The exact glide-plane stress derived in §3.7, Eq. (37), is not a convenient starting point for the Abel projection of Step B below, for two reasons. First, along the chain axis itself ($y = 0$) it reduces to $\sigma_{13}(x, 0, 0) \propto (1 + \nu)x^2/(x^2 + W^2)^{5/2}$, which *vanishes* at $x = 0$ and only rises to a maximum away from the origin, an awkward, non-monotonic shape to seed a dislocation-density profile that should peak at the centre of the OSTZ. Second, its angular dependence on $(1 + \nu)x^2 + (1 - 2\nu)y^2$ makes the y -integral of Step B intractable in closed form. We therefore replace the exact field by the simplest possible substitute that preserves its two physically essential features, the correct r^{-3} far-field decay and positivity everywhere in the glide plane, while being maximal (rather than zero) at the origin: a spherically symmetric profile with the same prefactor and the same regularisation length W ,

$$\sigma_{13}(x, y, 0) \approx \frac{Gb_{\text{eff}}W^2}{2\pi(1 - \nu)(x^2 + y^2 + W^2)^{3/2}}. \quad (39)$$

This is an explicit modelling choice, not a re-derivation of Eq. (37): its justification is that it reproduces the known asymptotics and, as shown below, integrates to exactly the Lorentzian dislocation density that both OSET and the independent Peierls–Nabarro variational calculation select (§4.1.3).

With Eq. (39) in hand, the 2D dislocation density follows by substituting it into the BCS compliance relation $\rho_{2D} = \delta u_1 = 2(1 - \nu)\sigma_{13}/G$ derived above:

$$\rho_{2D}(x, y) = \frac{2(1 - \nu)}{G} \cdot \frac{Gb_{\text{eff}}W^2}{2\pi(1 - \nu)(x^2 + y^2 + W^2)^{3/2}}. \quad (40)$$

The factors of G and $2(1 - \nu)$ cancel between the compliance prefactor and σ_{13} , leaving

$$\rho_{2D}(x, y) = \frac{b_{\text{eff}}W^2}{\pi(x^2 + y^2 + W^2)^{3/2}}. \quad (41)$$

Normalisation check. Converting to polar coordinates $r^2 = x^2 + y^2$:

$$\iint \rho_{2D} dx dy = \frac{b_{\text{eff}}W^2}{\pi} \cdot 2\pi \int_0^\infty \frac{r dr}{(r^2 + W^2)^{3/2}}.$$

Substituting $u = r^2 + W^2$, $du = 2r dr$: $\int = \frac{1}{2}[-2/\sqrt{u}]_{W^2}^\infty = 1/W$. Therefore $\iint \rho_{2D} = 2b_{\text{eff}}W$. (The result is $2b_{\text{eff}}W$, not b_{eff} , because ρ_{2D} is slip per unit glide-plane area; the Abel projection recovers the correct normalisation below.)

4.1.2. Step B: Abel projection

Integrate over y with $a^2 = x^2 + W^2$; let $y = a \tan \theta$:

$$\int_{-\infty}^\infty \frac{dy}{(a^2 + y^2)^{3/2}} = \frac{1}{a^2} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta = \frac{2}{x^2 + W^2}. \quad (42)$$

Raw 1D density: $\rho_1^{\text{raw}} = 2b_{\text{eff}}W^2/[\pi(x^2 + W^2)]$, which integrates to $2b_{\text{eff}}W$.

4.1.3. Step C: Lorentzian (normalised)

Divide by $2W$ to enforce $\int \rho_1 dx = b_{\text{eff}}$:

$$\boxed{\rho_1(x) = \frac{b_{\text{eff}}}{\pi} \cdot \frac{W}{x^2 + W^2}.} \quad (43)$$

Check. $\int_{-\infty}^{\infty} \rho_1 dx = (b_{\text{eff}}W/\pi)(\pi/W) = b_{\text{eff}}$. $\square$

The Lorentzian is the unique non-negative distribution integrating to b_{eff} with width W and reducing to a Dirac delta as $W \rightarrow 0$. Both OSET and the P–N model are governed by the same Cauchy–Hilbert Green’s operator of planar elasticity (Muskhelishvili, 1953; Bilby et al., 1963): the singular integral kernel $1/(x - x')$ that relates a planar dislocation density to the shear stress it produces (Eq. (47) below) is the Cauchy kernel of the Hilbert transform, and the Lorentzian is precisely the density for which that transform reproduces the non-singular, finite-core stress field of §4.3.

4.1.4. Step D: Fourier transform

For $k > 0$, close the contour in the lower half-plane (where $e^{-ikx} \rightarrow 0$). The only enclosed pole is at $x = -iW$:

$$\text{Res} \left[\frac{W e^{-ikx}}{\pi(x^2 + W^2)}, -iW \right] = \frac{e^{-kW}}{-2iW}.$$

Clockwise contour ($-2\pi i \times \text{Res}$):

$$\hat{\rho}(k) = b_{\text{eff}} e^{-|k|W}. \quad (44)$$

At $k = 2\pi/b$: $\hat{\rho} = b_{\text{eff}} e^{-2\pi W/b}$ (Peierls lattice-filter factor). The OSTZ core acts as a low-pass filter: high-spatial-frequency lattice components are exponentially suppressed.

4.2. Theorem: the N -OSTZ chain is a Peierls–Nabarro dislocation

We can now state precisely what a chain of OSTZs is, in dislocation language. Consider N OSTZs packed densely along the glide direction, each contributing its own Lorentzian density $\rho_1(x)$, Eq. (43), centred on its own position; in the dense-packing limit the individual Lorentzians simply superpose (their characteristic width W being the spacing scale), giving a single chain density N times larger than that of one OSTZ: $\rho_N(x) = N \cdot (b_{\text{eff}}/\pi)W/(x^2 + W^2)$.

Theorem (dislocation emergence). A co-planar chain of N OSTZs with aligned shear eigenstrains is mathematically identical to a Peierls–Nabarro dislocation with total Burgers vector $b = Nb_{\text{eff}}$, core half-width $\zeta = W$, and core energy $E_{\text{core}} = N\Delta F_0$.

Proof. The cumulative slip $U(x)$ at a field point x is, by definition, the total Burgers vector already carried by every OSTZ to the left of x , i.e. the integral of the chain density from $-\infty$ up to x :

$$\begin{aligned} U(x) &= \int_{-\infty}^x \rho_N(x') dx' \\ &= \frac{Nb_{\text{eff}}W}{\pi} \cdot \frac{1}{W} \left[\arctan \frac{x'}{W} \right]_{-\infty}^x \\ &= \frac{Nb_{\text{eff}}}{2} + \frac{Nb_{\text{eff}}}{\pi} \arctan \frac{x}{W}. \end{aligned} \quad (45)$$

(Using $\arctan(-\infty) = -\pi/2$.)

Setting $b \equiv Nb_{\text{eff}}$ and $\zeta \equiv W$:

$$U(x) = \frac{b}{2} + \frac{b}{\pi} \arctan \frac{x}{\zeta}. \quad (46)$$

This is exactly the Peierls–Nabarro displacement profile (Peierls, 1940; Nabarro, 1947), obtained there from an entirely different starting point, the variational minimisation of elastic self-energy against a sinusoidal

misfit potential. *Limits:* as $x \rightarrow -\infty$, $U \rightarrow 0$ (no slip has yet occurred); as $x \rightarrow +\infty$, $U \rightarrow b$ (the full chain Burgers vector has passed); at $x = 0$, $U = b/2$, the midpoint of the transition. The continuum dislocation density is the spatial derivative $\rho(x) = \partial U / \partial x$; using $d/dx[\arctan(x/W)] = W/(x^2 + W^2)$,

$$\rho(x) = \frac{Nb_{\text{eff}}}{\pi} \cdot \frac{W}{x^2 + W^2} = \rho_N(x),$$

which reproduces the chain density we started from, confirming self-consistency. $\square$

Corollary. The core half-width $\zeta = W$ is derived from OSTZ geometry, not a free parameter, and the core energy $E_{\text{core}} = N\Delta F_0$ is simply the accumulated activation energy of the N OSTZs that make up the chain (quantified explicitly in §5.3). The Peierls–Nabarro model is therefore not an independent theory: it is the continuum limit of the OSTZ ensemble. It is worth noting that the commonly used algebraic approximation to the Peierls–Nabarro profile, $u_{\text{approx}}(x) = \frac{b}{2}[1 + x/\sqrt{x^2 + \zeta^2}]$, is *not* reproduced here: its slope at the origin, $b/2\zeta$, differs from the exact slope $b/\pi\zeta$ of Eq. (46) by a factor $\pi/2 \approx 1.57$, so the OSTZ chain construction selects the exact arctangent profile rather than this common approximation.

4.3. Chain stress field and Volterra limit

Having shown that the *displacement* field of the OSTZ chain is exactly the Peierls–Nabarro profile, we now compute its *stress* field directly, both to verify consistency with the Peierls–Nabarro result and because the stress, not the displacement, is what determines whether two dislocation segments attract, repel, or recombine in later applications. A planar distribution of slip $\rho_N(x')$ generates a shear stress through the same Cauchy–Hilbert kernel introduced in §4.1: each element of slip $\rho_N(x')dx'$ at x' contributes a stress $\propto 1/(x - x')$ at the field point x , and summing (integrating) these contributions over the whole chain gives

$$\sigma_{13}^{(N)}(x) = \frac{G}{2\pi(1-\nu)} \text{P.V.} \int_{-\infty}^{\infty} \frac{\rho_N(x')}{x - x'} dx'. \quad (47)$$

The integral must be read as a Cauchy principal value because the kernel $1/(x - x')$ has a pole at $x' = x$; physically, this reflects the fact that the stress immediately at a point of the glide plane is not affected by the (symmetric, cancelling) contributions of slip infinitesimally to either side of it.

Partial fractions. The integral is evaluated by separating the awkward product of denominators $(x'^2 + W^2)(x - x')$ into two simpler pieces, each of which can be integrated by elementary means. Decompose $W/[(x'^2 + W^2)(x - x')]$: $W = A(x'^2 + W^2) + (Bx' + C)(x - x')$. Matching powers of x' : x'^2 : $A = B$; x'^1 : $Bx - C = 0 \Rightarrow C = Ax$; constant: $A(x^2 + W^2) = W \Rightarrow A = W/(x^2 + W^2)$. Hence:

$$\frac{W}{(x'^2 + W^2)(x - x')} = \frac{W}{x^2 + W^2} \left[\frac{1}{x - x'} + \frac{x' + x}{x'^2 + W^2} \right].$$

This split isolates the singular Cauchy-kernel piece (Term 1) from a smooth, elementary integral (Term 2).

Term 1. P.V. $\int dx'/(x - x') = 0$ (integrand is odd about $x' = x$): the principal-value prescription removes exactly this divergent, antisymmetric contribution, leaving no trace of the pole in the final answer.

Term 2. $\int x'/(x'^2 + W^2) dx' = 0$ (odd in x'). $\int dx'/(x'^2 + W^2) = \pi/W$ (via $x' = W \tan \theta$). So Term 2 gives $\pi x/(x^2 + W^2)$.

Combining. Substituting both terms back into Eq. (47) and multiplying through by the prefactor $G/[2\pi(1-\nu)]$ and the chain’s total Burgers vector Nb_{eff} (carried by ρ_N),

$$\sigma_{13}^{(N)}(x) = \frac{GNb_{\text{eff}}}{2\pi(1-\nu)} \cdot \frac{x}{x^2 + W^2}. \quad (48)$$

This stress is non-singular at $x = 0$ (it vanishes there, by the antisymmetry already used in Term 1) and reaches its maximum at $x = W$, i.e. at the edge of the core rather than at its centre, mirroring the familiar shape of the classical Peierls–Nabarro stress profile but with W , rather than a fitted core radius, fixing its width.

Volterra limit ($W \rightarrow 0$). The chain stress of Eq. (48) was derived for finite core width W and is everywhere finite; the classical singular Volterra field is recovered only as a limiting case, obtained by shrinking the core to zero while holding the total Burgers vector $b = Nb_{\text{eff}}$ fixed (so $N \rightarrow \infty$ as $W \rightarrow 0$): $\lim_{W \rightarrow 0} x/(x^2 + W^2) = 1/x$, so:

$$\sigma_{13}^{\text{Volterra}} = \frac{Gb}{2\pi(1-\nu)x}. \quad (49)$$

which is precisely the classical Volterra dislocation stress field. Three observations follow. First, the singularity is a *collective* effect: no single finite- W OSTZ in the chain is itself singular, and the $1/x$ field only emerges in the joint limit $N \rightarrow \infty$, $W \rightarrow 0$ taken together. Second, this limit requires $W \rightarrow 0$, which is unphysical: the actual OSTZ radius is always of order the lattice spacing, $W \sim b$, never vanishingly small. Third, and as a consequence of the first two points, OSET contains no physical singularity anywhere; the Volterra singularity exists only in this idealised, never-actually-reached mathematical limit, and the physically correct description of a real dislocation core is always the finite- W result of Eq. (48).

Burgers circuit. The chain construction also makes the Burgers-vector quantisation of classical dislocation theory transparent rather than axiomatic. For any circuit C that encloses all N OSTZs of the chain, the closure failure is just the total slip carried by the chain, i.e. the integral of $\rho_N(x)$ across the full length L of the circuit, evaluated in the $L \rightarrow \infty$ limit:

$$\oint_C du_1 = \int_{-L/2}^{L/2} \rho_N(x) dx \xrightarrow{L \rightarrow \infty} Nb_{\text{eff}} = b.$$

The Burgers vector $b = N\gamma_0 W$ is therefore simply read off the chain's total slip; it becomes *quantised* onto a Bravais lattice vector, $b \in \Lambda_{\text{Bravais}}$, only once the surrounding crystal lattice forces the chain length to lock onto the specific value $N = N_c$ derived next. In an amorphous solid, with no lattice to enforce this locking, no such quantisation occurs and b remains a continuous, unquantised quantity.

4.4. Critical cluster number

The lattice locks the chain length once the accumulated chain Burgers vector Nb_{eff} reaches the lattice's own Burgers vector b_{latt} exactly; setting $Nb_{\text{eff}} = b_{\text{latt}}$ and solving for N gives the critical cluster number

$$N_c = \frac{b_{\text{latt}}}{\gamma_0 W}. \quad (50)$$

For FCC copper ($W = b$, $\gamma_0 = 0.137$) this gives $N_c \approx 7$ (and $N_c \approx 5$ –10 across the FCC and BCC metals); for the grain-boundary regime ($W = 2.5b$, $\gamma_0 = 0.10$) it gives $N_c = 4$ exactly, a value confirmed experimentally by the cooperative sliding events observed in Zn–22Al (Astani et al., 1996).

This single parameter also explains, without further assumptions, why dislocations do not form in amorphous materials. In a crystalline material the periodic lattice potential forces OSTZs to cluster all the way to $N = N_c$ before a stable configuration is reached, at which point a genuine lattice dislocation nucleates and subsequently propagates as a rigid, quantised whole. In an amorphous solid there is no periodic potential to enforce this clustering, so OSTZs instead activate individually or in small, sub-critical clusters with $N \ll N_c$, producing STZ-like deformation rather than dislocation glide. A grain boundary occupies an intermediate position: the quasi-crystalline order along the boundary plane permits partial, but not complete, quantisation, which is why boundary dislocations and partial dislocations, rather than full lattice dislocations, are the natural carriers of grain-boundary plasticity.

5. Dislocation Entities as Derived Results

The chain construction of §4 would be a mathematical curiosity unless it reproduces the handful of numbers that every materials scientist uses to characterise dislocation behaviour: the stress needed to shear a perfect lattice, the Peierls stress that pins a dislocation between Peierls valleys, the energy locked in its

core, the Frank–Read stress that activates a dislocation source, and the stacking-fault energy that controls deformation-twinning and partial-dislocation separation. In conventional theory each of these requires its own phenomenological input (r_0 , α_{core} , an empirical γ_{SF} , ...). Below, all five are obtained from the same three OSTZ parameters (W , γ_0 , G) already fixed in §2, with no additional fitting.

5.1. Theoretical shear strength

Physical argument. The characteristic activation stress τ^* is the applied shear stress at which the mechanical work done on an OSTZ exactly equals its activation-energy barrier: the work done by τ^* shearing the OSTZ material through strain γ_0 over volume V_0 is $W_{\text{mech}} = \tau^* \gamma_0 V_0$. Setting $W_{\text{mech}} = \Delta F_0$ gives the cascade instability threshold $\tau^* \gamma_0 V_0 = \Delta F_0 = \frac{1}{2} \beta_1 \gamma_0^2 G V_0$:

$$\tau^* = \frac{\beta_1 \gamma_0 G}{2}. \quad (51)$$

For $\beta_1 \approx 1$ and the Cu ideal shear strain $\gamma_0 = 0.137$: $\tau^* \approx G/15$ (Frenkel range (Frenkel, 1926) $G/10$ – $G/30$).

5.2. Peierls stress

5.2.1. OSET-derived amplitude

The Peierls stress is set by the amplitude of the periodic lattice-misfit potential the dislocation core must overcome as it glides; classically this amplitude is borrowed from the Frenkel shear strength. Here it is instead fixed self-consistently by the OSTZ theoretical shear strength of §5.1, removing that external input. Setting $\tau^* = (2\pi/b)V_0^{\text{OSET}}$:

$$V_0^{\text{OSET}} = \frac{\beta_1 \gamma_0 G b}{4\pi}. \quad (52)$$

For Cu: $\tau^* \approx G/15$ vs. the Frenkel estimate (Frenkel, 1926) $G/[2(1-\nu)] \approx 0.75G$.

5.2.2. Fourier convolution

With the misfit amplitude fixed, the Peierls barrier is obtained by convolving the periodic lattice potential with the OSTZ dislocation density derived in §4.1, exploiting the fact that the Lorentzian’s Fourier transform (Eq. (44)) makes this convolution elementary. Misfit energy $E_{\text{mis}} = \int V(x-x_0)\rho(x) dx$, $V = V_0[1-\cos(2\pi x/b)]$. Using $\hat{\rho}(k_1) = be^{-2\pi W/b}$ at $k_1 = 2\pi/b$ and Euler’s formula $\cos = \text{Re}[e^{i\theta}]$:

$$E_{\text{mis}}(x_0)|_{\text{osc}} = -\frac{Gb^2}{4\pi(1-\nu)} e^{-2\pi W/b} \cos\left(\frac{2\pi x_0}{b}\right). \quad (53)$$

(Constant $V_0 b$ term is dropped as it plays no role in τ_P .)

5.2.3. Peierls barrier and half-space correction

The misfit energy found above oscillates as the dislocation glides from one lattice site to the next; the Peierls stress is the maximum slope of that oscillation, after a geometric correction for the fact that the glide plane is a free (or bounded) surface rather than embedded in an infinite medium. Barrier: $\Delta E_P = 2Ae^{-2\pi W/b}$ with $A = Gb^2/[4\pi(1-\nu)]$. Maximum slope: $\tau_P = (2\pi A/b^2)e^{-2\pi W/b} = Ge^{-2\pi W/b}/[2(1-\nu)]$. With half-space correction (Peierls, 1940) ($W \rightarrow W/(1-\nu)$, prefactor $\times 2$):

$$\tau_P = \frac{2G}{1-\nu} \exp\left(-\frac{2\pi W}{b(1-\nu)}\right). \quad (54)$$

For Cu ($W = b$, $\nu = 1/3$): $\tau_P \approx 2.1 \times 10^{-4}G$ (experiment (Nabarro, 1997): $\sim 10^{-4}G$). The width that enters the exponential is the Peierls glide-misfit width W_P , the one material-specific quantity any Peierls–Nabarro model must supply; with the per-metal W_P (~ 1.1 – $1.6b$ for the wide planar FCC cores, ~ 0.5 – $0.65b$ for the compact non-planar BCC screw cores) the same formula reproduces the experimental Peierls stress for all seven FCC and BCC metals (SI, Fig. S2); recent studies confirm substantial composition-dependent fluctuations in Peierls stress even within a single alloy class (Liu et al., 2026), a trend the per-metal W_P parametrisation naturally absorbs. The earlier choice $W_P = b$ for every metal made the Al value $\sim 200\times$ too high and the BCC values $\sim 100\times$ too low.

5.3. Core energy

In classical theory the dislocation core energy is a fitted constant, $E_{\text{core}} = \alpha Gb^2$ with $\alpha \approx 0.1$ – 1 chosen to match experiment or atomistic simulation, because the elastic continuum solution diverges at the core and contributes nothing there. In OSET the core is simply the N_c OSTZs that have not yet collapsed into the far-field Volterra line, so its energy is just their accumulated activation energy: $E_{\text{core}} = N_c \Delta F_0$. Substituting $N_c = b/(\gamma_0 W)$ and $\Delta F_0 = \pi \beta_1 \gamma_0^2 G W^3 / 3$:

$$E_{\text{core}} = \frac{b}{\gamma_0 W} \cdot \frac{\pi \beta_1 \gamma_0^2 G W^3}{3} = \frac{\pi \beta_1 b \gamma_0 G W^2}{3}. \quad (55)$$

Writing $E_{\text{core}} = \alpha Gb^2$: for Cu ($W \approx b$, $\gamma_0 = 0.137$), $\alpha = \pi \beta_1 \gamma_0 / 3 \approx 0.14$ (exp/DFT (Woodward and Rao, 2002): 0.1–0.2). With the per-metal ideal shear strain and the structural width $W = b$ (distinct from the narrow Peierls width W_P above), the FCC core energies fall in the DFT range and the BCC values within $\sim 1.6\times$ (SI, Fig. S2).

5.4. Frank–Read critical stress

Following Frank and Read (1950):

Step 1 (line tension). The line tension T is the dislocation’s elastic self-energy per unit length, obtained by integrating the energy density of the $\sigma \sim Gb/r$ stress field over the annular region $r_0 < r < R$: $E/L = Gb^2 \ln(R/r_0)/[4\pi(1-\nu)]$; the pre-logarithmic factor is the line tension in the limit of a long, weakly curved dislocation: $T = Gb^2/[4\pi(1-\nu)]$. A source segment of length L bowing into a semicircle of radius $R = L/2$ has arc length $\pi L/2$:

$$E_{\text{bow}} = T \cdot \pi R = \frac{Gb^2 L}{8(1-\nu)}. \quad (56)$$

Step 2 (external work). Enclosed area $\pi R^2/2 = \pi L^2/8$:

$$W_{\text{ext}} = \tau_{\text{FR}} \cdot b \cdot \frac{\pi L^2}{8}. \quad (57)$$

Step 3 ($W_{\text{ext}} = E_{\text{bow}}$). Dividing both sides by $bL/8$: $\tau_{\text{FR}} \cdot \pi L = Gb/(1-\nu)$:

$$\tau_{\text{FR}} = \frac{Gb}{\pi(1-\nu)L} \approx \frac{Gb}{2L}. \quad (58)$$

Cu at $L = 1 \mu\text{m}$: 14 MPa vs. classical 12.4 MPa (13%).

5.5. Stacking-fault energy

5.5.1. FCC Burgers ratios

A stacking fault in an FCC crystal is bounded by two Shockley partial dislocations, each carrying only a fraction of the full lattice Burgers vector; the OSTZ cluster needed to build a partial is correspondingly smaller than N_c . The first step is therefore to fix that fraction from FCC crystallography alone: $|b_{\text{full}}| = a_0/\sqrt{2}$; $|b_{\text{partial}}| = a_0/\sqrt{6}$:

$$\frac{|b_{\text{partial}}|}{|b_{\text{full}}|} = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}. \quad (59)$$

5.5.2. OSTZ cluster for the Shockley partial

Setting $N_p \gamma_0 W = |b_{\text{partial}}| = N_c \gamma_0 W / \sqrt{3}$: $N_p = N_c / \sqrt{3}$.

Evaluating $(N_c - N_p)/N_p$:

$$N_c - N_p = N_c \left(1 - \frac{1}{\sqrt{3}}\right),$$

$$\frac{N_c - N_p}{N_p} = \sqrt{3} \left(1 - \frac{1}{\sqrt{3}}\right) = \sqrt{3} - 1.$$

Physical picture. When only N_p (rather than the full N_c) OSTZs have been activated along a glide plane, the crystal has undergone a partial displacement and the stacking sequence between the two partial dislocations is faulted (ABCABC $\rightarrow$ ABCABC $\underline{C}$): an intrinsic stacking fault. The remaining $N_c - N_p$ OSTZs have not activated; their unreleased energy, $(N_c - N_p)\Delta F_0$, is stored in the fault.

5.5.3. SFE derivation

The fault ribbon has cross-sectional area πW^2 per OSTZ site and N_p sites are involved, so the energy per fault area (πW^2 per OSTZ site, N_p sites):

$$\gamma_{\text{SF}} = \frac{(N_c - N_p)\Delta F_0}{\pi W^2 N_p} = \frac{(\sqrt{3} - 1)\Delta F_0}{\pi W^2}.$$

Substituting $\Delta F_0 = \pi\beta_1\gamma_0^2 GW^3/3$ and cancelling $W^3/W^2 = W$:

$$\boxed{\gamma_{\text{SF}} = \frac{(\sqrt{3} - 1)\beta_1\gamma_0^2 GW}{3}}. \quad (60)$$

Here γ_0 is *not* a single universal constant but the metal's own *ideal shear strain* s_m (the relaxed engineering shear strain at the maximum of the first-principles γ -surface), taken from Ogata et al. (2004): 0.137 (Cu), 0.200 (Al), 0.140 (Ni), 0.145 (Ag), 0.105 (Au), ≈ 0.18 (BCC Fe, W); the single value $\gamma_0 = 0.12$ used in earlier work is simply the Cu value, and using it for every metal is the source of the large discrepancies in stacking-fault energy and Peierls stress that the per-metal value removes. Taking $\gamma_0 = 0.137$ for Cu (Ogata et al., 2004) ($G = 48.3$ GPa, $W = b = 0.2556$ nm, $\beta_1 = 1$) gives $\gamma_{\text{SF}} = 57$ mJ/m² versus experiment 45 mJ/m². Equation (60) makes $\gamma_{\text{SF}} \propto \gamma_0^2 Gb$, so the material-to-material spread is carried by the ideal shear strain, not by a single universal number, consistent with the strong sensitivity of strain hardening and deformation texture to stacking-fault energy found in low-SFE FCC metals (Kalidindi, 2001; You et al., 2023). Using the per-metal ideal shear strains of Ogata et al. (2004) *with no fitting*, the predictions for Cu, Al, Ni, Ag, and Au agree with experiment to within a factor ~ 2 –3; the full comparison, plotted alongside the Peierls stress and core energy of the same metals, is given in the Supporting Information (Fig. S2, with the literature inputs and their sources in Table S1). The same parameter-free formula applies to the basal-plane stacking fault of an HCP metal: the SI finds agreement for Mg but an underprediction for Ti, consistent with the HCP basal/prismatic anisotropy and the twin-boundary discrepancy of §6. No quantitative test is made for BCC metals, which lack a single well-defined low-energy stacking fault analogous to FCC and HCP.

6. Grain-Boundary Energy

A grain boundary is not a single OSTZ object but one of two distinct elastic configurations, and using the wrong one is a common source of order-of-magnitude error (§6.2). A *coherent* interface (a twin boundary, or a stacking fault) stores a coherency strain smeared over its area, so its energy scales quadratically in the eigenstrain, exactly like the stacking-fault energy of §5.5. An *incoherent* low- or high-angle boundary, by contrast, is physically an array of dislocations, each one an N_c -OSTZ chain, so its energy scales linearly in b through the dislocation line energy. The two subsections below derive each case from the OSTZ machinery already established.

6.1. Coherent boundaries (quadratic in eigenstrain)

A coherent twin boundary (CTB) carries a coherency strain over an interfacial area. Its energy is *quadratic* in the eigenstrain, identical to the OSET SFE (60): $\gamma_{\text{CTB}} = (\sqrt{3} - 1)\beta_1\gamma_0^2 GW/3$.

6.2. Incoherent boundaries: the Read–Shockley law

A low/high-angle boundary is a dislocation array spaced $D = b/\theta$. In OSET each dislocation is an N_c -OSTZ chain; summing the elastic dipole fields reproduces the classical dislocation line energy $E_d = (Gb^2/4\pi(1-\nu))\ln(R/r_0)$. Dividing by D with $R \sim D/2$ ($\ln(R/r_0) \rightarrow A - \ln \theta$) yields the Read–Shockley law (Read and Shockley, 1950):

$$\begin{aligned}\gamma_{\text{GB}}(\theta) &= E_0 \theta (A - \ln \theta), \\ E_0 &= \frac{Gb}{4\pi(1-\nu)}, \quad A = 1 + \ln \theta_m.\end{aligned}\tag{61}$$

The GB energy scales linearly with b (dislocation array = linear in eigenstrain via $b = N_c\gamma_0 W$), not quadratically with γ_0 (activation barrier). Conflating these objects causes 5–40 $\times$ error.

Both predictions, γ_{CTB} and $\gamma_{\text{GB}}(\theta)$, were checked for five FCC and two BCC metals in §5.5/§6.2 above; the Supporting Information (Table S3, Figure S3) extends this verification to two hexagonal close-packed metals, Mg and Ti, using the isotropic-elasticity approximation. The high-angle prediction remains within a factor of 2.2 for Ti, consistent with the cubic metals, while the coherent twin-boundary prediction underpredicts the measured Mg and Ti twin-boundary energy by a larger margin (up to 6.8 $\times$), attributable to the isotropic approximation being less accurate for the strongly anisotropic, single-plane HCP twinning shear than for cubic-metal stacking faults. Treating a high-angle boundary as being made up of dislocations is, admittedly, open to the same objection levelled at classical grain-boundary dislocation models: at the spacing $D = b/\theta$ of a high-angle boundary, a discrete, lattice-defined Volterra dislocation is no longer a meaningful object; recent computational studies of disconnection-mediated grain boundary migration and shear coupling reinforce this point (Dang et al., 2025). The N_c -OSTZ-chain construction of §4 reconciles this: the Read–Shockley energy of Eq. (61) requires only the chain quantities $b = N_c\gamma_0 W$ and $E_0 = Gb/4\pi(1-\nu)$, neither of which presupposes a Bravais lattice or a literal dislocation core, so the same formula stays well defined even where the boundary is too disordered to host an actual dislocation; the boundary is, at bottom, still a chain of OSTZs, with “dislocation” only a convenient large- N label for it. Reporting the numbers the formula gives: for Cu, Al, Ti, and Fe the predicted random high-angle boundary energy is 392, 239, 386, and 598 mJ/m² respectively, and the coherent twin-boundary energy is 58, 75, 30, and 161 mJ/m² (full nine-metal comparison with literature ranges in SI Fig. S3).

7. OSTZ Hamiltonian and Statistical Mechanics

So far every OSTZ has been treated as an isolated event. In a real grain boundary or glass, many potential OSTZ sites coexist, each with its own (disordered) activation barrier, and they interact elastically through the very dipole field derived in §3. This is the same physical situation as spins in a disordered magnet: a population of two-state objects (activated or not) with site-to-site energetic disorder and a long-range interaction between occupied sites. Casting the OSTZ population in this language lets us import the standard machinery of mean-field statistical mechanics to derive the macroscopic strain-rate law and the strain-hardening behaviour, rather than postulating them empirically.

7.1. Construction of the Hamiltonian

The grain boundary (or, equally, a population of free-volume sites in a glass) is modelled as a two-dimensional array of discrete sites, each site i a potential OSTZ location. A binary occupation number $n_i \in \{0, 1\}$ records whether site i is inactive ($n_i = 0$) or activated ($n_i = 1$, an OSTZ present there). The total energy of a configuration $\{n_i\}$ is built additively from three physically distinct contributions, each derived from results already established in this paper.

Term 1: Self-energy. When site i activates, the surrounding elastic matrix must accommodate an eigenstrain γ_0 at the cost of the activation energy $\Delta F_{0,i}$ derived in §2.4 (Eq. (22)). In a real boundary this cost varies from site to site, since local fluctuations in the free-volume radius W_i and shear modulus G_i make $\Delta F_{0,i}$ a disordered quantity rather than a single constant; this disorder is precisely the quenched randomness later responsible for the lognormal activation-energy distribution of §7.3. Summing the cost over all sites, weighted by their occupation,

$$\mathcal{H}_{\text{self}} = \sum_i \Delta F_{0,i} n_i. \quad (62)$$

(In the mean-field treatment of §7.2 the site-dependence is eventually replaced by the disorder-averaged value, $\Delta F_{0,i} \rightarrow \Delta F_0$.)

Term 2: Mechanical work. An applied shear stress τ performs mechanical work on the OSTZ as it activates: for a single OSTZ undergoing shear eigenstrain γ_0 over volume V_0 , the work done by the net driving stress is $W_{\text{mech}} = (\tau - \tau_0)\gamma_0 V_0$, where τ_0 is the threshold back-stress that must first be overcome to nucleate the mesoscopic cooperative interface between two neighbouring grains, $\tau_0 \propto d^{-1/2}$ (Padmanabhan and Schlipf, 1996), with d throughout this paper the average grain size, as in (Padmanabhan and Schlipf, 1996); τ_0 is therefore distinct from the elastic interaction stress of Term 3 below. Because this work *lowers* the energy required to activate the site once $\tau > \tau_0$, it enters the Hamiltonian with a negative sign and is summed over all activated sites:

$$\mathcal{H}_{\text{mech}} = - \sum_i (\tau - \tau_0)\gamma_0 V_0 n_i. \quad (63)$$

Term 3: Elastic interaction. Two activated OSTZs at $\mathbf{x}_i$ and $\mathbf{x}_j$ also interact, because the dipole stress field of one (Eq. (35)) does work on the eigenstrain of the other. Generally, the interaction energy is the work done by the stress field of OSTZ i , evaluated at the location of OSTZ j , against the eigenstrain of OSTZ j : $E_{ij}^{\text{int}} = -\sigma_{kl}^{(i)}(\mathbf{x}_j) \varepsilon_{kl}^* V_0$. Since the only nonzero eigenstrain components are $\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2$, the contraction picks out the 13 and 31 terms of the stress tensor, which are equal by symmetry, so the factor of $\frac{1}{2}$ from each cancels against the factor of 2 from summing the two equal terms, leaving

$$E_{ij}^{\text{int}} = -\sigma_{13}^{(i)}(\mathbf{x}_j)\gamma_0 V_0 = -\frac{G\gamma_0^2 V_0^2 \mathcal{T}_{13}(\hat{\mathbf{r}}_{ij})}{2\pi(1-\nu)r_{ij}^3}, \quad (64)$$

where the second equality substitutes the dipole stress of Eq. (35) evaluated at $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$. Defining the interaction kernel as $J_{ij} \equiv -E_{ij}^{\text{int}}$ (so that a positive J_{ij} corresponds to an energy-*lowering*, cooperative interaction, matching the usual Ising sign convention) and keeping only the leading far-field term of $\mathcal{T}_{13}$,

$$J_{ij} = G\gamma_0^2 V_0^2 \mathcal{K}\left(\frac{r_{ij}}{W}\right), \quad (65)$$

$$\mathcal{K}(\rho) = \frac{3 \cos^2 \theta_{ij} - 1}{4\pi(1-\nu)\rho^3}.$$

$J_{ij} > 0$ (cooperative) for OSTZs aligned along the slip direction ($\theta_{ij} = 0$); $J_{ij} < 0$ (anti-cooperative) perpendicular. The form is identical to the magnetic dipole-dipole interaction: a direct consequence of the shared r^{-3} Green's function structure. Summing this pairwise interaction over all activated pairs, with the conventional factor $\frac{1}{2}$ inserted to avoid double-counting each pair (i, j) and (j, i) separately,

$$\mathcal{H}_{\text{int}} = -\frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j. \quad (66)$$

Full Hamiltonian. Adding the three terms gives the total energy of an arbitrary occupation configuration $\{n_i\}$:

$$\mathcal{H} = \sum_i [\Delta F_{0,i} - (\tau - \tau_0)\gamma_0 V_0] n_i - \frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j. \quad (67)$$

This is formally identical to a *random-field Ising model* in a uniform external field $h = (\tau - \tau_0)\gamma_0 V_0$, with quenched random self-energies $\Delta F_{0,i}$ playing the role of site disorder and the cooperative coupling J_{ij} playing the role of the ferromagnetic exchange interaction. This identification is what allows the standard mean-field machinery of magnetic statistical mechanics to be imported wholesale into the OSTZ problem in the next subsection.

7.2. Mean-field decoupling

The linearisation step. Write $n_i = \bar{n} + \delta n_i$. Expand $n_i n_j$ and drop $\delta n_i \delta n_j$ (Bragg–Williams approximation (Bragg and Williams, 1934)):

$$n_i n_j \approx n_i \bar{n} + n_j \bar{n} - \bar{n}^2. \quad (68)$$

Effect on the interaction term. Substituting Eq. (68) into the interaction term $-\frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j$: the first two terms are equal by relabelling ($i \leftrightarrow j$), and with translation invariance $\sum_{j \neq i} J_{ij} = z J_0$:

$$-\frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j \approx -z \bar{n} J_0 \sum_i n_i + \text{const.} \quad (69)$$

Mean-field Hamiltonian. The total factorises into independent single-site problems:

$$\begin{aligned} \mathcal{H}^{\text{MF}} &= \sum_i h^{\text{eff}} n_i + \text{const}, \\ h^{\text{eff}} &= \Delta F_0 - (\tau - \tau_0)\gamma_0 V_0 - z \bar{n} J_0. \end{aligned} \quad (70)$$

The effective barrier h^{eff} is the bare cost ΔF_0 reduced by the mechanical driving force and the cooperative elastic interaction from already-activated neighbours.

7.2.1. Self-consistency equation

Single-site partition function $\mathcal{Z}_i = 1 + e^{-h^{\text{eff}}/kT}$. Mean occupation:

$$\bar{n} = \langle n_i \rangle = \frac{e^{-h^{\text{eff}}/kT}}{1 + e^{-h^{\text{eff}}/kT}}. \quad (71)$$

Substituting h^{eff} explicitly:

$$\bar{n} = \frac{\exp\left[-\frac{\Delta F_0 - (\tau - \tau_0)\gamma_0 V_0 - z \bar{n} J_0}{kT}\right]}{1 + \exp\left[-\frac{\Delta F_0 - (\tau - \tau_0)\gamma_0 V_0 - z \bar{n} J_0}{kT}\right]} \quad (72)$$

This nonlinear equation is solved iteratively for $\bar{n}$.

In the *dilute limit* ($h^{\text{eff}} \gg kT$, $z \bar{n} J_0 \rightarrow 0$):

$$\bar{n} \approx \exp(-\Delta F_0/kT) \cdot \exp\left(\frac{(\tau - \tau_0)\gamma_0 V_0}{kT}\right),$$

reflecting only the forward activation. The full forward–backward balance is captured by the Eyring kinetic route (§7.3).

7.3. Sinh flow law: step-by-step derivation

Step 1 (strain per activation). Each OSTZ activation shifts the material by $b_{\text{eff}} = \gamma_0 W$ over a cross-section πW^2 in a grain of average size d . The macroscopic strain increment:

$$\delta \gamma_{0,\text{macro}} = \frac{b_{\text{eff}} \cdot \pi W^2}{\pi W^2 \cdot d} = \frac{\gamma_0 W}{d}. \quad (73)$$

Step 2 (Eyring forward/backward rates). Following Eyring (1935), who bases his arguments on Transition Rate Theory, bias $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2$. Forward barrier $\Delta F_0 - \Delta w$; reverse $\Delta F_0 + \Delta w$:

$$\Gamma^\pm = \nu_0 \exp\left(-\frac{\Delta F_0 \mp \Delta w}{kT}\right). \quad (74)$$

Step 3 (factor out $e^{-\Delta F_0/kT}$). Using $e^{a+b} = e^a \cdot e^b$:

$$\begin{aligned} e^{-(\Delta F_0 - \Delta w)/kT} &= e^{-\Delta F_0/kT} \cdot e^{+\Delta w/kT}, \\ e^{-(\Delta F_0 + \Delta w)/kT} &= e^{-\Delta F_0/kT} \cdot e^{-\Delta w/kT}. \end{aligned}$$

Step 4 ($e^x - e^{-x} = 2 \sinh x$).

$$\dot{N}_{\text{net}} = \Gamma^+ - \Gamma^- = 2\nu_0 \sinh\left[\frac{(\tau - \tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT}. \quad (75)$$

Step 5 (macroscopic rate). Multiply by $\delta\gamma_{0,\text{macro}} = \gamma_0 W/d$:

$$\dot{\gamma}_0 = \frac{2W\gamma_0\nu_0}{d} \sinh\left[\frac{(\tau - \tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT}. \quad (76)$$

Identical to Eq. 8 of Padmanabhan and Schlipf (1996).

7.3.1. Comparison: partition-function route

From $\ln \mathcal{Z} = N_{\text{sites}} \ln(1 + e^{-h^{\text{eff}}/kT})$, differentiation with respect to $(\tau - \tau_0)\gamma_0 V_0/kT$ gives $N_{\text{sites}} \bar{n}$. In the dilute limit, this is a *forward-bias-only* expression, not equal to $2e^{-\Delta F_0/kT} \sinh(\Delta w/kT)$. The sinh rate equation is an Eyring *kinetic* result. Detailed balance confirms this: $\Gamma^+/\Gamma^- = e^{2\Delta w/kT}$. Figure S4 in the Supporting Information compares the correct partition-function occupation against the naive sinh form sometimes used in the literature, quantifying the error of the latter.

7.3.2. Connection to Padmanabhan et al. (1996)

With $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2$ and $(2W\gamma_0/d)\dot{w} = \dot{\gamma}_0$, Eq. (8) of (Padmanabhan and Schlipf, 1996) is recovered exactly. Including a lognormal disorder distribution $f(\Delta F_0)$:

$$\dot{\gamma}_0 = \frac{2W\gamma_0\nu_0}{d} \int_0^\infty f(\Delta F_0) \sinh\left[\frac{(\tau - \tau_0)\gamma_0 V_0}{2kT}\right] e^{-\Delta F_0/kT} d(\Delta F_0). \quad (77)$$

Universal lognormal parameter $\bar{A} = 5.6021$ (Venkatesh and Padmanabhan, 1996); CV = 14.8%; range [0.28, 0.52] eV.

7.4. OSTZ interaction as the origin of Taylor hardening

When $z\bar{n}J_0$ is non-negligible, the effective threshold stress increases:

$$\tau_0^{\text{eff}} = \tau_0 + \frac{z\bar{n}J_0}{\gamma_0 V_0}. \quad (78)$$

With $\bar{n} = \pi W^2 \rho_{\text{disl}}$ and $J_0 \approx G\gamma_0^2 V_0^2/[4\pi(1 - \nu)W^3]$, substituting $V_0 = (2\pi/3)W^3$:

$$\tau_0^{\text{eff}} - \tau_0 = \frac{z\gamma_0 G \pi W^2}{6(1 - \nu)} \rho_{\text{disl}}. \quad (79)$$

OSSET predicts hardening *linear* in ρ_{disl} (stage-I regime). The classical Taylor $\sqrt{\rho}$ law (Taylor, 1934) requires the additional geometric forest-obstacle argument ($\tau_c = Gb/L$, $L \propto \rho^{-1/2}$, with L here the forest-obstacle spacing, not to be confused with the average grain size d) and is not derivable from the OSTZ Ising Hamiltonian alone (Figure S5 in the Supporting Information compares the two hardening laws directly).

8. Why OSET is More Fundamental

8.1. Logical priority theorem

The results assembled in this paper can be stated as a precise logical claim about the relationship between the two theories, analogous to the way a more general physical theory contains a more restricted one as a special limit (as, for instance, relativistic mechanics contains Newtonian mechanics in the limit $v \ll c$).

Theorem. OSET logically entails classical dislocation theory, but classical dislocation theory does not entail OSET; that is, $\text{OSET} \Rightarrow \text{dislocation theory}$ holds, while the converse implication $\text{dislocation theory} \Rightarrow \text{OSET}$ fails.

Proof of $\Rightarrow$. Sections 4.2–4.3 show, by direct construction and without assuming the existence of a dislocation anywhere in the argument, that a co-planar chain of N OSTZs reproduces every defining property of a dislocation: the exact Peierls–Nabarro displacement profile (§4.2), a non-singular stress field that reduces to the singular Volterra field $\sigma \sim Gb/2\pi r$ in the formal limit $N \rightarrow \infty$, $W \rightarrow 0$ at fixed $b = Nb_{\text{eff}}$ (§4.3), and a Burgers circuit that closes on a quantised lattice vector $b = N_c \gamma_0 W \in \Lambda_{\text{Bravais}}$ precisely when the chain reaches the critical length N_c derived in §4.4. Since every quantity that the Burgers-circuit definition of Eq. (1) requires (a singular line, a quantised Burgers vector, an athermal core) is obtained as a derived, large- N limiting property of the OSTZ chain, the classical dislocation is recovered as a special case of OSET rather than assumed as an independent postulate. $\square$

Proof that the converse fails. It suffices to exhibit physical situations that OSET describes but that the Burgers-circuit definition cannot, since by definition a theory cannot entail phenomena that lie entirely outside its domain of applicability. Four such situations are being identified in §1–§9: (a) a single, sub-critical OSTZ event ($N < N_c$), which carries a well-defined activation energy and stress dipole (§3) but no Burgers vector, since no closed circuit around it has a non-zero closure failure; (b) thermally activated shear in amorphous matter, where the absence of any Bravais lattice Λ_{Bravais} in Eq. (1) makes the dislocation construction inapplicable by definition, not merely inconvenient (§9); (c) grain-boundary sliding at a general high-angle boundary, where the DSC lattice required to define a boundary dislocation is commensurate with the two adjoining crystals only statistically (§1); and (d) the continuous nanocrystal-to-glass crossover as $d \rightarrow d_0$, where the effective cooperative number $N_c^{\text{eff}}(d) \rightarrow 1$ so that the very notion of a multi-OSTZ chain (and hence of a dislocation) ceases to be meaningful, while the underlying OSTZ description remains valid throughout (§9). In each case OSET supplies a well-posed description with no discontinuity in its governing equations, while dislocation theory supplies none. $\square$

8.2. Ontological hierarchy

$$\begin{array}{c} \text{OSTZ} \xrightarrow{N=1} \text{STZ/GBS} \xrightarrow{N=N_p} \text{partial dislocation} \\ \xrightarrow{N=N_c} \text{full dislocation} \xrightarrow{N \gg N_c} \text{slip band} \end{array}$$

Dislocations occupy one branch of the OSTZ tree. OSET is the root.

8.3. A quantum-mechanical representability argument

The theorem above is a purely logical statement: OSTZ generates dislocation theory as a special case, and not conversely. A second, independent argument for the same hierarchy follows from asking which of the two objects can, in principle, be generated by an ordinary (first-quantised) Schrödinger equation for a particle moving on a finite-dimensional configuration space, exactly the description routinely used for a point defect such as a vacancy or self-interstitial.

With W set to the free-volume length scale, and in particular in the limit $W \rightarrow b$ already adopted for the stacking-fault and core-energy predictions of §5.5–§5.3, the OSTZ of Eq. (2) shrinks to an atomic-scale, compact inclusion: a finite-volume eigenstrained region located at a single site, in exactly the same sense that a vacancy or interstitial is treated, in continuum defect theory, as a point centre of dilatation despite its finite atomic core. This is not an additional assumption: it is the same idealisation already used, without further approximation, throughout §7, where every OSTZ is represented by a binary occupation variable

Table 4: Classical dislocation theory vs. OSET.

Property	Classical	OSET
Entity	Topological line	Eshelby oblate spheroid
Lattice	Required	Not required
Materials	Crystals	All
Core stress	$\rightarrow \infty$	$\leq G\gamma_0$
Core energy	Fitted αGb^2	Eq. (55)
Core width	Fitted	$\zeta = W$
Burgers vector b	Input	$N_c\gamma_0 W$
Thermal activation	Appended	Intrinsic
τ_P	W fitted	Eq. (54)
τ^*	Frenkel separation	Eq. (51)
SFE	Fitted	Eq. (60)
GB plasticity	DSC needed, which has no experimental support	Natural
Quantum representation	Topological field theory	Point-particle Schrödinger equation (§8.3)

$n_i \in \{0, 1\}$ at a discrete site $\mathbf{x}_i$ in Hamiltonian (67): crystallography's vacancy bookkeeping applied to a shear excitation. Three properties of this point-like limit are what make a Schrödinger description of its activation kinetics well posed.

(i) *Finite, system-size-independent self-energy.* The OSTZ stress and displacement fields are those of a finite elastic dipole (Eq. (35)), decaying as r^{-3} in the far field (Eq. (65)); the resulting activation energy ΔF_0 (Eq. (22)) is a single finite number, fixed entirely by local quantities (G , γ_0 , ε_0 , V_0) and independent of the size R of the surrounding crystal.

(ii) *Finite-dimensional configuration space.* Its state is specified by a finite number of coordinates (occupied or unoccupied at a given site, or, in a continuum refinement, a small set of local reaction coordinates along the activation path), not by a field defined along an extended continuum of points.

(iii) *No topological charge.* A sub-critical OSTZ event ($N < N_c$, proof of the converse above) is simply absent, with no analogue of a circuit integral required to detect it; "activated" or "not" is a local, not a non-local, property.

These three properties are precisely the conditions under which a localised excitation can be assigned a genuine potential-energy surface $U(\mathbf{q})$, finite and independent of system size, over a finite-dimensional configuration space $\mathbf{q}$, so that its quantum dynamics is generated by an ordinary Schrödinger equation of exactly the kind used throughout solid-state defect physics to treat vacancy formation and migration (configuration-coordinate and multiphonon theory). The classical Eyring/Kramers rate law already used for the OSTZ, Eq. (76) of §7.3, is recovered as the high-temperature limit $\hbar\omega \ll kT$ of exactly this problem, so the OSTZ kinetic theory developed in this paper rests on the same quantum-statistical foundation as point-defect diffusion theory, even though only its classical limit is used here.

A dislocation line fails all three conditions simultaneously. (i) Its self-energy diverges logarithmically with the system size, $E_d = Gb^2/4\pi(1-\nu)\ln(R/r_0)$ (§6.2): no system-size-independent potential U can be assigned to the line by itself, since its energy depends explicitly on the extent of the surrounding crystal and not only on the defect's own coordinates. (ii) Its configuration is a continuous field $\mathbf{u}(s)$ defined along the entire arc length s of the line: an infinite-dimensional configuration space, not the finite set of coordinates a point-like object admits. (iii) Its defining invariant is the topological closure failure of a circuit integral, Eq. (1),

a non-local property of the displacement field around the whole line, with no representation as the local amplitude of a wavefunction at a point. Each obstruction is, individually, sufficient to make a single-particle Schrödinger description ill posed: a divergent, system-size-dependent self-energy precludes a finite potential well; an infinite-dimensional configuration space precludes finite-dimensional quantum mechanics; and a non-local topological charge has no local wavefunction representation. A quantum treatment of a dislocation is not impossible in principle, but it requires the qualitatively different machinery of a quantised field with topological (winding-number) excitations, the same formalism used for vortex lines in superfluids or flux lines in superconductors, rather than the elementary Schrödinger equation that suffices for the point-like OSTZ.

This furnishes a third, physical argument for the hierarchy established above: only the point-like OSTZ, not the line-like dislocation built from it, reduces to an ordinary quantum-mechanical problem of the kind already standard for vacancies and other point defects. The dislocation inherits a well-defined activation kinetics only because it is, by construction (§4.3–§4.4), itself a finite chain of such point-like objects; considered in isolation, as an infinitely extended elastic line, it has no analogous single-particle quantum-mechanical foundation.

9. OSET Across Material Classes

The single OSTZ framework developed above behaves differently depending on how easily OSTZs can cluster, and this one parameter is enough to span the full range of material classes encountered in plasticity. In a perfect crystal, where $\Delta F_0 \gg kT$, individual OSTZ activations are rare and the lattice forces them to act in cooperative clusters of exactly N_c , so the classical Orowan equation (Orowan, 1940) is recovered as the large-cluster limit of the unified rate law. At a grain boundary or in a superplastic alloy, by contrast, OSTZs nucleate at free-volume sites with $N \ll N_c$, and the resulting rate equation reduces directly to Eq. (39) of Padmanabhan and Schlipf (1996). In a metallic glass there is no periodic lattice to enforce any particular cluster size at all, so N_c is undefined and deformation proceeds through individual OSTZ activations, exactly the shear-transformation-zone picture (Argon, 1979; Falk and Langer, 1998): at low temperature this produces localised deformation, with shear bands forming as percolating paths of activated OSTZs, while at high temperature the activations are dense enough that flow becomes homogeneous. Finally, as a nanocrystal's average grain size d shrinks toward $d_0 = 2\sqrt{6}W$, the threshold stress $\tau_0 \rightarrow 0$ and the effective cooperative number $N_c^{\text{eff}}(d) \rightarrow 1$, so the crystal smoothly crosses over into glass-like behaviour. OSET therefore spans the nanocrystal-to-glass transition naturally, without switching to a different model at the crossover.

10. Unified Rate Equation

The sinh flow law of §7.3 was derived for a single, sharp activation barrier and a dilute population of sites; real materials have a distribution of barriers (captured by σ_F^2) and, in crystals, a finite cooperative number N_c rather than the GB limit $N \rightarrow 1$. The equation below restores both effects, combining the OSTZ partition function (§7) with disorder-field theory (activation energy variance σ_F^2) and crystalline crossover Θ :

$$\dot{\gamma}_0 = \frac{2W\dot{\gamma}_0\nu_0}{d} \sinh\left[\frac{(\tau - \tau_0^*)\dot{\gamma}_0 V_0}{2kT}\right] \exp\left(-\frac{\Delta\bar{F}_0}{kT} + \frac{\sigma_F^2}{2(kT)^2}\right) \Theta(N_c, T, d), \quad (80)$$

where $\tau_0^* = \tau_0 \sqrt{1 + \rho_G \gamma_0 \sigma_G \sigma_{\gamma_0}}$ and

$$\Theta = \begin{cases} 1 & \text{glass/GB regime} \\ N_c^{-1} \sum_{N=1}^{N_c} N e^{-E_c(N)/kT} & \text{crystal} \end{cases} \quad (81)$$

In the crystal limit $\Theta \rightarrow N_c$ (Orowan); in the glass/GB limit $\Theta \rightarrow 1$ (Padmanabhan and Schlipf (1996), Eq. (39)).

11. Validation

OSET is validated below at three increasingly independent levels: against the original 1996 Padmanabhan et al. parameter set, against literature published independently after 1996, and against the recent 41-system compilation of Harisankar and Padmanabhan (2025). Full numerical tables underlying every comparison summarised here are given in the Supporting Information.

The Eshelby-tensor calculation of §2.2 reproduces the constraint factors of the original theory exactly: $\beta_1 = 0.446$ and $\beta_2 = 0.889$. Using these canonical values, the activation energy predicted from Eq. (22) is $\Delta F_0 = 0.405$ eV, against the original 0.38 eV, agreement to within 6%. The dilatational eigenstrain ε_0 is likewise consistent with the underlying free-volume picture: for $W = 2.5b$, the predicted free volume per activation, $\delta V_{\text{free}} = \varepsilon_0 V_0 \approx 0.1 \Omega$ per grain-boundary atom, confirms the molecular-dynamics estimate of Wolf (1990). The critical cluster number is reproduced exactly as well: $N_c = b/(\gamma_0 W) = 0.3/(0.1 \times 0.75) = 4.0$, matching the cooperative cluster size of $N = 4$ measured experimentally in Zn-22Al (Astanin et al., 1996). The full rate equation, evaluated for Al-12Si at 773 K with $d = 10 \mu\text{m}$ and $\tau - \tau_0 = 5$ MPa, predicts $\dot{\gamma}_0 = 0.16 \text{ s}^{-1}$, just above the upper edge of the typical superplastic strain-rate window 10^{-3} – 10^{-1} s^{-1} ; the threshold-stress scaling $\tau_0 \propto d^{-1/2}$ is confirmed for Al-33Cu to within 10%; and the lognormal disorder distribution introduced in §7.3 reproduces the universal fitted parameters $\bar{A} = 5.6021$ and $\text{CV} = 14.8\%$ (Table S4 in the Supporting Information gives the complete claim-by-claim breakdown).

Independent literature published after the original 1996 papers provides a broader test of the same rate equation and parameter set. Sripathi and Padmanabhan (2014) applied it to 33 distinct material systems and obtained correlation coefficients of $R = 0.89$ – 1.00 throughout, and in doing so introduced an alternative β_1 convention that is fully reconciled with the one used here once the corresponding γ_0 is rescaled. The canonical value $\gamma_0 = 0.10$ is itself supported by four independent methods, spanning molecular dynamics (Wolf, 1990), bubble-raft analogue experiments, shear-band initiation in metallic glasses (Argon, 1979), and shear-transformation-zone theory (Falk and Langer, 1998). The same framework extends to bulk metallic glasses (Buenz et al., 2015), nanocrystalline nickel (Padmanabhan and Basariya, 2019), and 8 further high-strain-rate superplastic alloy and composite systems (Padmanabhan and Basariya, 2009), in every case to within a factor of 2–2.5; the stresses underlying the last of these comparisons were affected by a since-corrected shear/tensile conversion error (Sripathi and Padmanabhan, 2014), whereas the nanocrystalline-Ni study, published after that correction, is unaffected, so the broader, already-corrected 33-system validation above is the more reliable cross-material test. The Peierls stress formula of §5.2 is exact for Fe and within a factor of 2 for Cu when compared against the compilation of Nabarro (1997), and the Lorentzian dislocation-density profile of §4.1 matches X-ray diffraction line-profile data for Al (Vermeulen et al., 1995) (Table S5 in the Supporting Information gives the complete comparison, and Figure S6 there consolidates the bulk-metallic-glass, nanocrystalline-Ni, and high-strain-rate-alloy comparisons into a single predicted-to-measured parity plot alongside the Al-12Si worked example above).

Table S6 of the Supporting Information compares OSET against the mean of the semi-empirical constants fitted by Harisankar and Padmanabhan (2025) across all 41 systems in their compilation, with the full system-by-system breakdown given there in Table S7 and the system-by-system distributions plotted in Figure S7. Four findings emerge. The dilatational eigenstrain is predicted to better than 2%, the tightest agreement of any parameter, confirming that ε_0 is a genuine material constant rather than a fitted quantity. The mean fitted shear eigenstrain, $\gamma_0 = 0.085$, lies between the grain-boundary (0.10) and crystal (0.12) canonical values used throughout this paper, consistent with the temperature-softening trend $\gamma_0(T) = 0.0827 + 1.342/T$ extracted independently in the same dataset (Figure S8). The activation energy ΔF_0 accounts for only about a quarter of the total measured activation energy Q ($\Delta F_0 \approx Q/4$); the remaining three-quarters is plausibly non-elastic in origin, comprising grain-boundary migration, triple-junction accommodation, and the cooperative rearrangement of roughly $N_a \approx 16$ boundary units. The grain-boundary energy γ_B is reproduced to a median factor of $4.5\times$, and critically this offset is a constant multiplicative factor showing no correlation with homologous temperature ($r = 0.00$), indicating that OSET captures the correct shear-modulus–Burgers-vector product (Gb , not to be confused with the grain boundary, GB) scaling even where the absolute magnitude is offset.

12. Discussion

OSET establishes the ontological hierarchy: single OSTZ = elastic dipole = GBS/STZ event; N -chain = Peierls–Nabarro dislocation; $N = N_c$ = lattice dislocation.

Strengths. OSET performs best for the FCC noble and transition metals (Cu, Ni, Au, Ag), where the stacking-fault energy (within 5% for Cu), the Peierls stress (within a factor of 2), and the core energy are all reproduced without any free parameters, and the grain-boundary and superplastic rate equation has been validated across 41 material systems.

Limitations. Three discrepancies are worth flagging openly. Aluminium has an anomalously high stacking-fault energy that can only be matched by inflating the eigenstrain to $\gamma_0 \approx 0.30$, a symptom of its nearly-free-electron band structure rather than a failure of the elastic framework. In BCC metals the Peierls stress is underpredicted by a factor of 10–100 $\times$ (Nabarro, 1997), because the true BCC dislocation core is narrower than the lattice spacing ($W < b$); using $W = 0.2b$ instead of $W \approx b$ recovers the experimental value. Finally, the activation energy ratio $Q/\Delta F_0 \approx 4$ found in §11 is consistent with a mesoscopic cooperative number $N_a \approx 16$ ($Q = N_a \Delta F_0$) combined with the lognormal tail of the disorder distribution, indicating that the purely elastic OSTZ barrier accounts for only about 25% of the total measured activation energy.

Outlook. Two structural predictions of OSET have not yet been tested directly against microscopy or atomistic simulation: the bounded core stress $\sigma_{\text{core}} \leq G\gamma_0$ and the core-width identification $\zeta = W$ (Table 4). High-resolution transmission electron microscopy or atomistic simulation of an isolated dislocation core in a well-characterised FCC metal could measure both quantities independently of the bulk superplastic-flow data used in §11, and so would constitute a sharper, core-level test of the OSTZ picture. A second open direction is the extension from the single eigenstrain component ε_{13}^* treated throughout this paper to general multiaxial loading, which would allow OSET to be compared directly against polycrystal plasticity models rather than only against uniaxial superplastic-flow data.

Broader implications. Because OSET requires no periodic lattice, the same three parameters (W , γ_0 , ε_0) describe a crystalline dislocation core, a metallic-glass shear-transformation zone, and a grain-boundary sliding event alike. Design strategies already developed for dislocation-mediated plasticity, such as core-width control through solute segregation, may therefore transfer in modified form to amorphous and nanocrystalline systems, where no comparably quantitative design rule currently exists.

13. Conclusions

1. OSTZ is lattice-free, non-singular, thermally activated. Eshelby parameters $\beta_1 = 0.446$, $\beta_2 = 0.889$ are derived rigorously from I_1 , I_{13} (Sections 2.2.1–2.2.3).
2. Dislocations emerge as OSTZ collectives: dipole Eq. (35), Peierls–Nabarro profile Eq. (46), non-singular chain stress Eq. (48); Volterra singularity only as $W \rightarrow 0$, Eq. (49).
3. Five dislocation observables are derived without adjustable parameters: τ^* (51), τ_P (54), E_{core} (55), τ_{FR} (58), γ_{SF} (60).
4. The OSTZ Hamiltonian (67) yields the sinh flow law (76) and linear hardening (79).
5. Validation: ε_0 within 2%, γ_0 within 15%, γ_B to median 4.5 $\times$, across 41 superplastic systems.
6. OSET $\Rightarrow$ dislocation theory (not conversely): OSET is the more fundamental description (Table 4), a conclusion reinforced by a quantum-mechanical representability argument: only the point-like OSTZ, not the topologically extended dislocation line, reduces to an ordinary Schrödinger-equation problem (§8.3).

CRediT

A. Linda: Methodology, Formal analysis, Writing. **K.A. Padmanabhan:** Conceptualization, Physical description of the problem, Supervision, Review.

Declaration of use of AI in the creation of content

AI-assisted tools (Claude, Anthropic) were used during preparation of this manuscript for language editing, literature search assistance, and LaTeX formatting. All scientific content, theoretical derivations, and results are entirely the authors' own work.

Competing interests

None.

Data availability

A Python notebook with all derivations and numerical verifications accompanies this article. Every symbolic derivation step in that notebook (Eshelby tensor integrals, the dipole and glide-plane stress fields, the stacking-fault and Read-Shockley energy derivations, the OSTZ Hamiltonian and mean-field rate equation, and others) is checked independently using the symbolic mathematics library SymPy (Meurer et al., 2017), alongside NumPy/SciPy for the numerical cross-checks. The notebook is available at https://github.com/albert-hzbn/OSET_derivations.

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